Design and Development of an Effective Classifier for Medical Images based on Custom CNN

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students of Electronics and Communication Engineering Department, Vasavi College of Engineering in partial fulfilment of the requirement for the award of the degree of Bachelor of Engineering in Electronics and Communication Engineering is a record of the bona fide work carried out by them during the academic year 2025-2026. The result embodied in this project report has not been submitted to any other university or institute for the award of any degree

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ABSTRACT

Early detection of brain tumours is key to better treatment results, yet manual examination of MRIs can be tedious, energy-consuming, and prone to mistakes. Complexity of the images and individual differences between patients complicate the matter further. In light of this, the current research is aimed at creating a system for the automatic recognition of brain tumours using machine and deep learning algorithms.

At the initial stages of this work, standard machine learning algorithms, such as SVM, KNN, and Naïve Bayes, were used on pre-processed MRIs. Dimensionality reduction was achieved using feature extraction techniques, such as HOG and PCA. Nonetheless, such algorithms are heavily dependent on carefully selected features, which are unable to identify complex structures found in MRI images.

To eliminate these constraints, deep learning techniques have been employed, specifically fine-tuned ResNet18, EfficientNet-B0, MobileNetV2, and custom-made MobileNet. Notable amongst these efforts is the design of the Custom Convolutional Neural Network based on a fine-tuned ResNet18 model, increasing the automatic feature learning capability of the proposed brain tumour diagnosis framework. Prior to feeding the input to the network, the provided medical MRI data will be preprocessed using denoising, histogram equalization, K-Means Segmentation, and feature extraction algorithms.

The system under design is tasked with performing not only binary but also multiclass classifications. Therefore, besides distinguishing between tumour and non-tumour instances, it will be able to recognise the type of the existing brain lesion, namely glioma, meningioma, and pituitary tumours. Soft Voting is implemented in the system in an effort to improve classification accuracy by employing a combination of prediction results generated by different models.

The experiments performed prove that the proposed solution provides a significant improvement compared to the alternative traditional machine learning models with regard to both good accuracy rates , precision, and processing speed. The limitation of the system is in the need for a sufficient amount of high-quality labelled input.

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CHAPTER 1: INTRODUCTION

Brain tumours refer to a disease condition that can affect patients of any age group, and thus, early diagnosis is key for successful treatment. Normally, Magnetic Resonance Imaging (MRI) is used to observe the brain for the presence of tumours. However, it is quite difficult to conduct the examination using this method due to various factors, including the variation in tumour shape and size, among others. Moreover, since the process involves human analysis, the whole procedure becomes more dependent on the experience of the person doing the analysis, which may lead to mistakes due to fatigue and tiredness.

Therefore, there arises the need for developing an automatic system that can detect brain tumours. Traditionally, machine learning techniques such as SVM, KNN, and Naïve Bayes have been used in performing this analysis. Even though these techniques give satisfactory results in classification, they perform poorly since they depend on the extraction of manually computed features.

The method used in this project involves the use of an ensemble that uses both classical machine learning approaches and more modern deep learning techniques. Here, the dataset comprises of images which are classified either as a tumour image or not. In case of machine learning algorithms, the process of image preprocessing will involve the following activities: converting images to grayscale, noise reduction, segmenting of images, and finally principal component analysis (PCA) dimensionality reduction. Deep learning algorithms, on the other hand, will work on the resized images (224 × 224) and hence no need for manual feature extraction.

As a way of enhancing the performance of the ensemble, it is important to combine various deep learning algorithms such as ResNet18, ResNet50, MobileNetV2, DenseNet201, and EfficientNet-B0 and then apply a soft voting strategy on them. In addition to binary classification, the system is extended to multiclass classification, where tumours are categorized into specific types such as glioma, meningioma, and pituitary tumours. This makes the system more useful in real-world clinical scenarios, as different tumour types require different treatment strategies.

Additionally, in order to improve the interpretability aspect of the system, a local LLM will be added through the use of Ollama, thereby enabling the system to not only make predictions but also provide a clear and easy-to-understand explanation for its output.

In conclusion, the proposed system offers a viable solution for automatic brain tumour detection, as it leverages various methodologies including preprocessing, machine learning, deep learning, ensemble methods, and explainable AI in an attempt to lower diagnostic time and eliminate the occurrence of errors during the process.

1.1Motivation:

Nevertheless, contemporary methods used to diagnose brain tumours have certain disadvantages in relation to manual analysis of MRI images, which is a laborious task and involves high subjectivity and risk of errors. Therefore, there is an urgent need for the creation of special tools capable of aiding doctors in making more precise and faster decisions about the patient's health condition.

Moreover, most contemporary methods are limited in their capabilities, as they can give only the information about the disease being present or absent, whereas, from the practical point of view, what is needed is a classification of a particular type of brain tumour, namely, whether it is a glioma, meningioma, or pituitary tumour. One more disadvantage of the modern models is connected with their inability to detect complex spatial patterns in medical images.

Fortunately, the problem described above can be solved by using a deep learning model. Nevertheless, from the practical point of view, the problem is still topical in relation to the lack of a proper user interface and interpretability of the results.

1.2 Problem Statement:

The main difficulties associated with the classification of brain tumours by means of MR images are due to the complexity of the problem itself, as well as the image quality and image noise. On top of that, current methods of diagnosis use human

input, which makes the process very time consuming and subjective and leads to inconsistency in some cases.

Current machine learning solutions use manually created features, thus lacking ability to identify complex features in medical images, resulting in low accuracy rates. One additional issue is that current classifiers can perform binary classification (tumour/normal) only instead of the more complex multi-class classification.

Taking all of the aforementioned into account, one could argue that there is a necessity for creation of an automated system capable of addressing such problems as low efficiency, high time consumption, low accuracy rate, and the ability to perform multi-class classifications.

1.3 Objectives

- The main objective of this research is to develop an efficient and precise brain tumour classification algorithm from MRI data based on machine learning and deep learning algorithms.
- To conduct a literature review and analyse the datasets available, namely the Brain Tumour Classification Dataset (Tumour/No Tumour), Brain Cancer MRI Dataset (Glioma, Meningioma, Pituitary) by using machine learning and deep learning methods.
- To improve the quality of MRI images through the process of preprocessing that includes noise removal, histogram equalisation, and k-means clustering and feature extraction, including HOG and Principal component analysis.
- Implementing and comparing classic machine learning algorithms, SVM, KNN, and Naïve Bayes for binary and multi-class tumours.
- Using the Soft Voting Ensemble technique to increase the robustness of models using probabilistic output from several models for binary classification.
- Enhancing classification performance using deep learning models such as Custom CNN (based on ResNet) for binary classification and Custom

EfficientNet-B0, Mobile Net, and Custom Mobile Net for multi-classification.

- Comparing the efficiency of models based on accuracy, precision, and recall.
- Designing an end-to-end application in MATLAB using deep learning models coupled with a locally running language model called Ollama for automated tumour classification.

1.4 Dataset Description

The data used in brain tumour datasets employed in this research are made up of MRI images, which can either be used for binary classification or multiclass classifications. In the binary classification, the data is organized into two classes, namely tumour and non-tumour classes. Training data used for binary classification comprises 2,870 images of which 2,475 images correspond to tumour class while 395 images belong to non-tumour class, whereas the test set is made up of 394 images with 289 of tumour class and 105 non-tumour class images.

On the other hand, for multiclass classification, data is collected for identifying various types of tumours, which include glioma, meningioma, and pituitary tumours. The multiclass classification data is divided into train, validation, and testing data in the proportion of 70%, 15%, and 15% respectively. Glioma contains 648 images for training, 139 images for validation, and 139 images for testing, while Meningioma holds 656 training images, 141 validation images, and 141 testing images. Similarly, pituitary tumours have 631 images for training, 135 validation, and 135 testing images.

Images have been varied in terms of their resolution, brightness, contrast, and orientation, hence making the dataset more like the clinical setting. The process of converting images for classical machine learning is through transforming them into gray-scale images, followed by processing of the images by using noise reduction, histogram equalization, image segmentation, and extraction of features. On the other hand, in deep learning models, the resizing of images to be $224 \times 224 \times 3$ is carried out. This ensures that the model can learn patterns directly from the data.

Other methods like rotation, flipping, scaling, and translation of data are used for increasing data variance in order to improve accuracy. Therefore, the dataset can be considered as robust enough for training and testing machine learning and deep

learning models.

TABLE 1.4.1 Data set**:** Brain Tumour classification (Binary Classification)

	No Tumour	Tumour	Total
Train count	395	2475	2870
Test count	105	289	394

TABLE 1.4.2 Data set**:** Brain Cancer MRI (Multiclass Classification)

Tumour Class	Train count (70%)	Validation count (15%)	Test count (15%)
Glioma	648	139	4096
Meningioma	656	141	141
Pituitary	631	135	135

Software and Tools:

The preprocessing, feature extraction, model training, and classification processes in this project have been done in MATLAB R2023b. MATLAB software made it easier to process the images using various image processing techniques, design machine learning algorithms, and train deep learning models. With MATLAB software and its various tools such as App Designer, it was also possible to design a graphical user interface (GUI) for automated brain tumor classification.

1.6 OUTLINE OF THE THESIS

Chapter 1 gives the introduction, aim, motivation of the thesis

Chapter 2 gives the information about literature review

Chapter 3 discuss about various image preprocessing techniques

Chapter 4 gives the information about the implementation of machine learning classifiers

Chapter 5 discuss about Deep learning models and voting algorithms

Chapter 6 Discuss the results of the various models

Chapter 7 Conclusions

Chapter 8 Future Scope

Chapter 9 References

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The classification of brain tumors based on MRI has emerged as a crucial domain of study due to its significance in early detection and treatment. Various methods have been proposed throughout the years, ranging from classical machine learning to complex deep learning algorithms. There has also been considerable effort in enhancing the quality of images, feature extraction, and ensemble modeling to improve performance. The following discussion provides a critical review of significant works in this field.

- **Classical Machine Learning Approaches for Tumor Classification**
  However, during the initial phase of the research, most research works made use of machine learning methods like Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Naïve Bayes. For instance, Cheng et al. [1] utilized the SVM method for brain tumor classification using hand-crafted features derived from the MRI image. In another work, Zhang et al. [2] made use of Principal Component Analysis (PCA) with the SVM and KNN method to enhance classification speed. Despite achieving satisfactory performance, the major challenge in these approaches was their heavy reliance on feature extraction.

- **Importance of Preprocessing and Segmentation**
  Researchers have highlighted the importance of image preprocessing prior to classification. Various types of preprocessing techniques, such as noise filtering, histogram equalization, and normalization, are applied to enhance the quality of the image. Tustison et al. [3] proposed an enhanced bias correction technique that contributes towards generating homogeneous images from MR scans. Moreover, tumor region detection is carried out using the segmentation process, including K-means clustering algorithm, as explained by Selvaraj et al. [4].

- **Deep Learning for Improved Feature Learning**

Due to improvements in deep learning techniques, CNNs are now popularly used in classifying brain tumors. Such models can automatically extract features from the MRI images, thus eliminating the process of manually extracting features. According to Hossain et al. [6], ResNet-based models have proven successful in tumor classification due to their high accuracy, which is better than traditional methods. EfficientNet, introduced by Tan and Le [7], is also superior in terms of its architecture scaling optimization.

- **Role of Data Augmentation**

One of the most common problems in medical imaging is a lack of available data. To solve this problem, data augmentation methods like rotation, flipping, and rescaling can be used to augment the dataset. It was found by Shorten and Khoshgoftaar [8] that data augmentation contributes to better generalization and mitigates the problem of overfitting.

- **Ensemble Learning for better performance**

In the last few years, the use of combination modeling has become a very popular technique. Ensembles like the soft voting method take the predictions of all the models and make a final decision based on them. According to Zhou [9], these types of methods tend to reduce the shortcomings of each model and result in very consistent outcomes. Despite this, ensembles are mostly used as a supplemental tool.

- **Multiclass Classification in Medical Imaging**

Previous studies have concentrated more on binary classification, in which the objective is to distinguish the presence or absence of a tumor. But new research trends are concerned with classifying the tumors into various categories like glioma, meningioma, and pituitary. According to a study conducted by Cheng et al. [10], multiclass classification results in more information than binary classification. Deep neural networks can prove helpful in doing this work due to their effectiveness in distinguishing among classes.

- **Need for Explainable AI in healthcare**

Despite the high accuracy rate attained by most of these models, they are often hard to interpret because they tend to function like a "black box." This can pose a challenge when applying machine learning algorithms to medicine, where transparency is important. As pointed out by Samek et al. [11], explainable AI is critical in establishing trust in automated processes.

2.2 CONCLUSION

As can be inferred from the literature, there has been evolution in the area of classification of brain tumours from using simple machine learning algorithms to sophisticated deep learning techniques. Whereas the former provides a baseline approach, the latter ensures greater precision by allowing automatic feature extraction. In addition, the use of ensemble techniques makes the model more robust, and multiclass classification makes the approach more relevant in practical settings.

Table 2.2 Summary of Literature Survey

Ref No	Paper Title: Year of Publication	Implementation Techniques	Remarks
[16]	Design and development of an effective classifier for medical images based on machine learning and image segmentation (2024)	Machine Learning classifiers (SVM, KNN, Naïve Bayes), Image preprocessing techniques, Image segmentation (K-means / thresholding), Feature extraction methods	Improves classification accuracy by combining image segmentation with machine learning techniques. Segmentation helps in isolating regions of interest (tumour areas), leading to better feature extraction.
[18]	Automated Retinal Disease Classification Using Fine-Tuned ResNet50 (2025)	ResNet50, transfer learning, CNN-based feature extraction, image preprocessing, fine-tuning	Improves early disease detection accuracy; demonstrates strong performance using pretrained deep learning models.
[10]	Multiclass Brain Tumor Recognition Using CNN (2023)	CNN architecture, MRI classification, feature learning, softmax classifier	Accurately classifies multiple tumor types; improves diagnostic decision-making.

[8]	Skin Lesion Classification Using Pre-Trained DenseNet201 Deep Neural Network (2021)	DenseNet201, transfer learning, CNN-based classification, preprocessing	Achieves high accuracy; highlights efficiency of deep pretrained architectures
[6]	A survey on deep learning in medical image analysis (2017)	Review of CNNs, deep learning architectures, feature extraction methods	Provides foundational overview of deep learning techniques used in medical imaging.
[11]	Image Binary Classification of Coronary Artery Stenosis Based on ResNet18 and Transfer Learning (2023)	ResNet18, transfer learning, binary classification, preprocessing	Shows effectiveness of pretrained models in medical image classification tasks.
[7]	The significance of fruit weight as an indicator of watermelon maturity (2018)	Statistical analysis, feature-based evaluation, agricultural data modeling	Irrelevant to medical imaging; should be removed from literature survey for consistency.
[5]	Review of MRI-based brain tumour image segmentation using deep learning methods (2016)	Deep learning models (CNN), segmentation techniques, MRI analysis	Provides comprehensive review of segmentation approaches; useful for understanding DL-based methods.
[2]	The multimodal brain tumour image segmentation benchmark (BRATS) (2015)	Benchmark dataset, multimodal MRI analysis, segmentation evaluation	Provides standard dataset for comparing tumor segmentation methods.

[16].Almukhtar F.H., Kareem S.W., Khoshaba F.S., Design and development of an effective classifier for medical images based on machine learning and image segmentation, Egyptian Informatics Journal, Vol. 25, 100454, 2024. https://doi.org/10.1016/j.eij.2024.100454

CHAPTER 3: IMAGE PREPROCESSING TECHNIQUES

3.1 INTRODUCTION

The image preprocessing stage is vital because it ensures that the MR images can be used effectively in subsequent processing tasks. Since MR images have the potential of having noise and other irregularities in terms of intensity levels, it is necessary that they undergo preprocessing. This will enhance the quality of the images by making key features such as tumors more visible.

3.2 STEPS INVOLVED

3.2.1 Gaussian Elimination

The images obtained from MRI are first transformed to grayscale and later filtered and enhanced to remove any noise or artifacts present in the images. This improves the image quality and enables the identification of features within the image. The process of averaging intensities smooths out the boundaries of the tumours.

3.2.2 Histogram equalization

Histogram equalization is a common approach for preprocessing images in order to enhance their contrast and make them more visually clear. In the case of brain tumour recognition, the image quality directly influences the accuracy of detection of regions with abnormalities. Most of the MRI images are characterised by having low contrast, thus making it hard to differentiate the normal brain tissue from areas affected by tumours. Using histogram equalisation helps to solve this problem through redistributing frequent intensities in the image.

3.2.3 K-means Segmentation

K-Means Segmentation is one of the popular algorithms employed for unsupervised clustering in the field of image processing that enables an image to be split into regions of interest. It is mainly applied in medical imaging applications such as identifying brain tumors where K-Means is employed to demarcate the tumor regions from the normal brain tissues. This technique involves the clustering of

pixels into a predetermined number of clusters depending on their similarities, whereby pixels are assigned to the most similar clusters whose similarity is determined based on the average values (centroids). Together with other region-based techniques, K-Means segmentation forms an integral part of the tumor extraction process.

3.2.4 Feature Extraction through PCA (Principal Component Analysis)

Since medical images such as MRI scans are composed of many pixels, the end result is very high-dimensional data, e.g. 256x256 image translates to more than 65,000 dimensions. This type of data processing is quite complex and sometimes includes irrelevant information. For this purpose, the Principal Component Analysis (PCA) technique is used to reduce the dimension of the data. In doing so, PCA maps the original data onto new axes, where each feature becomes an uncorrelated component carrying most significant information and variation from the original data. Finally, from the tumor regions that have been isolated using segmentation, important features are extracted such as shape, texture, and intensity.

3.3 MRI Characteristics

The brain MRIs are made up of pixel intensity values in high dimensions, denoted as:

$X = R\hat{}(H \times W)$ (3.1)

where H refers to the height of the image while W refers to the width of the image. The dimensionality of a particular brain MRI image is:

$d = H$ (3.2)

Such high-dimensional input spaces are associated with the problem of high dimensionality that makes probability and distance-based classical models highly unstable owing to sparsity in the feature space.

Brain MRIs are also influenced by a number of elements such as noise modelled as: n(i,j)~"Rician Distribution",

Intensity inhomogeneity and tumour boundaries with highly irregular shapes.

3.4 Preprocessing Theory for Classical ML

3.4.1 Normalization

In order to normalise the intensity values of MRI scans and eliminate variations among scanners, X_norm is calculated as follows:

$$X_"norm" = (X - \mu)/\sigma \quad (3.3)$$

where μ and σ are the mean and standard deviations, respectively.

3.4.2 Histogram Equalisation

The equation p(i) is used to represent the probability of the level i.

$$C(i) = \sum_(j = 0)\hat{}i\, p(j) \quad (3.4)$$

Whereas, the equalized value of pixel intensity can be determined through:

$$X' = (L - 1) \cdot C(X) \quad (3.5)$$

where L represents the total number of gray levels.

3.4.3 K-Means Segmentation

The K-means clustering algorithm separates the pixels into K clusters based on:

$$J = \sum_(i = 1)\hat{}N \sum_(k = 1)\hat{}K\, r_ik \parallel x_i - \mu_k \parallel \hat{}2 \quad (3.6)$$

Where xi is the vector value of the pixels, μk is the centre point of the cluster, and rik = 1 when pixel i belongs to cluster k.

3.4.4 PCA (Dimensionality Reduction)

Considering the centred matrix:

$$\tilde{X} = X - \bar{X}, \quad (3.7)$$

The covariance matrix is calculated as:

$$C = 1/(n - 1) * \tilde{X}\hat{}T\tilde{X}. \quad (3.8)$$

Principal Components are defined by eigen decomposition as follows:

$$Cv_i = \lambda_i\, v_i. \quad (3.9)$$

A lower-dimensional representation Z can be derived as follows:

$$Z = \tilde{X}V_k. \quad (3.10)$$

Table 3.4.4. Features before and after PCA

Dataset	Images	Images Size	Features before PCA	Features after PCA	Dimensionality Reduction
Training	2870	64x64	4096	1245	69.6%
Testing	394	64x64	4096	1245	69.6%

3.4.5 Data Augmentation

Transformation performed on the image:

T(x) = {x_rot, x_flip, x_zoom, x_shift, x_bright}.

The effective size of the training dataset will be:

n' = n × |T|. (3.11)

Data augmentation adds more diversity and helps avoid overfitting in CNNs.

3.4.6 Histogram of Oriented Gradients (HOG)(used in Multiclass classification)

HOG extracts the information of the image in terms of structural and shape information, particularly edge detection and boundary detection rather than relying on pixel-based information. In other words, HOG does not directly work on the intensities of pixels; instead, it takes into consideration the gradients.

The first step involves the calculation of the horizontal and vertical gradient of the image:

$G_x = \partial I/\partial x, G_y = \partial I/\partial y$ (3.12)

The magnitude and angle at each pixel can be calculated based on the gradient:

"Magnitude: " $M(x,y) = \sqrt{}(G_x\text{^}2 + G_y\text{^}2\ )$ (3.13)

"Orientation: " $\theta(x,y) =$ 〖tan 〗 ^(-1) $(G_y/G_x\ )$ (3.14)

Now the image will be segmented into small square regions (also referred to as cells in your algorithm). The sizes of the cells vary from one application to another; however, you can use an 8×8 matrix for your implementation. Then we construct a histogram for each cell depending on the gradient direction of each pixel (for instance, from 0°-180°), and the magnitude of the gradient will affect the

contribution of each pixel within the histogram.

To make sure that these histograms are stable, we normalize the histograms by using larger regions called blocks. In other words, the features are calculated on blocks. Finally, all these histograms will be combined.

3.4.7 Z-Score Normalisation (Multi-class classification)

However, after HOG feature extraction, the range of the values will differ. Some features might contain large values while others might contain smaller values. This will be harmful to the model. Z-score normalisation will be done in this case.

The features will be normalised as follows:

$$Z = (X - \mu)/\sigma \quad (3.15)$$

where:

X=Original feature value

μ= Mean of that particular feature

σ= Standard Deviation

This will lead to the following result for the dataset:

Mean = 0

Standard Deviation = 1

Normalisation will help to bring all the features into one scale. It is particularly useful for algorithms such as SVM and KNN, where distance measures are used.

3.5 Summary

The presented algorithm represents an efficient method of brain tumour classification due to the combination of image pre-processing, image segmentation, feature extraction, and machine learning. Every phase of the algorithm increases the input data quality and classification efficiency. The structured extraction and normalisation of features allow attaining high accuracy for multiclass classification tasks, and dimensionality reduction helps to make the model more efficient. Finally, the classification model represents a robust solution for tumour recognition.

CHAPTER 4: Machine Learning Classifiers

4.1 INTRODUCTION:

Machine learning classifiers play a crucial role in the automated analysis of MRI brain images for tumour detection. In this work, preprocessed MRI images are used as input to classification algorithms such as Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Naïve Bayes. These classifiers learn patterns from extracted features and classify images into tumour and non-tumour categories. The overall methodology involves preprocessing, feature extraction, and classification to improve accuracy and reliability.

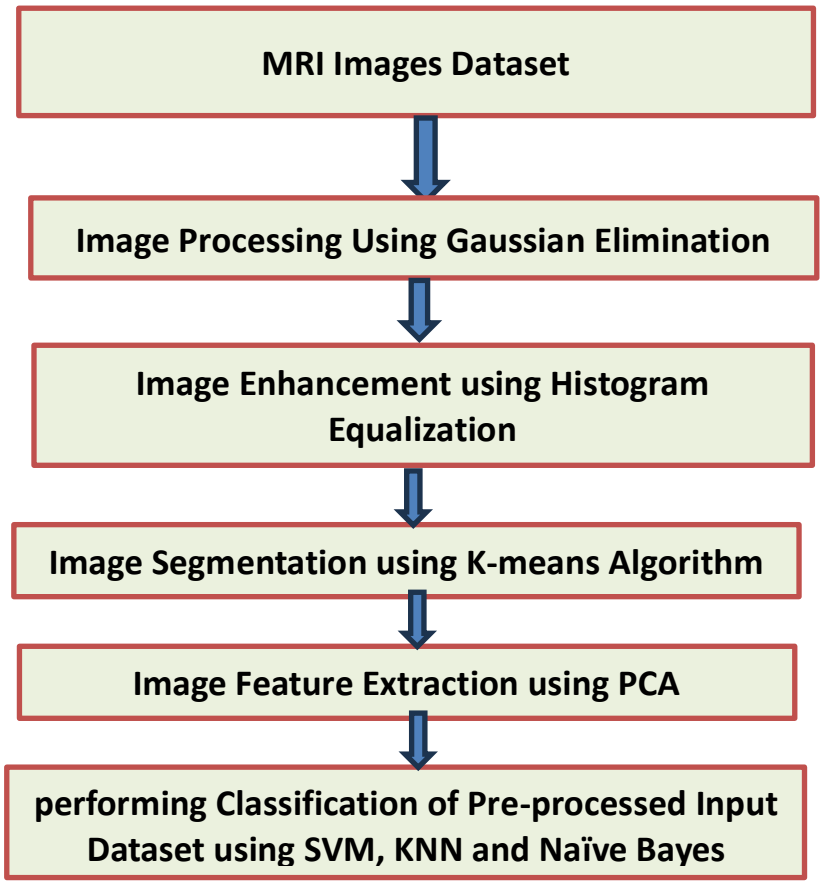


Fig 4.1. A methodology for classifying and segmenting MRI Images

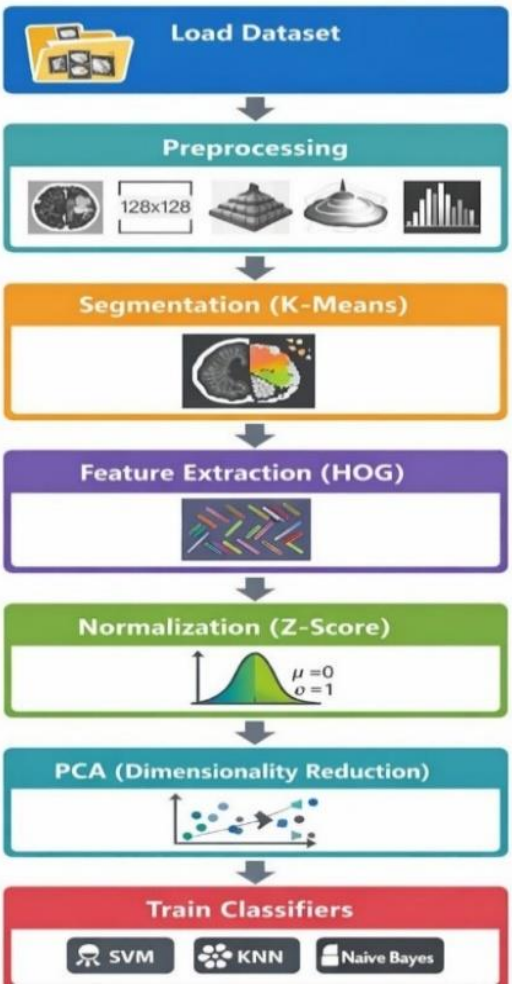


Fig 4.2 Methodology for Multiclass classification

4.2 Support Vector Machine

4.2.1 Overview of Support Vector Machine (SVM)

Support Vector Machine (SVM) is the name of a supervised machine learning algorithm, which tries to maximise the margin by finding the optimal decision boundary that can classify the MRI brain images in several categories.

However, for non-linearly separable data sets, SVM uses kernels like RBF and Polynomial Kernels to project the data points into higher-dimensional space for further analysis. These kernel functions help in better classification by taking advantage of the intricate features of MRI images analysed by image processing techniques.

4.2.2 Important Features of Support Vector Machine

Maximisation of Margin:

The margin concept used by SVM ensures that there is an optimal boundary obtained in the MRIs of the brains.

Versatility for Different Types of Data: It is important to state that SVM can handle both linear and non-linear datasets. Kernel functions are known for extracting features from complex MRIs, hence increasing the accuracy of tumour detection.

Avoiding Overfitting:

The support vector machine is based only on the most relevant training examples referred to as support vectors, hence reducing the likelihood of over-fitting. This characteristic is important for its use in medical imaging because labeled examples may not always be sufficient.

Handling of Non-linear Datasets Using Kernels:

Kernels enable SVMs to map the non-linear distribution of MRIs into a high-dimensional space for the purpose of distinguishing between the different types of tumours.

Effective in High-Dimensional Spaces:

SVM is efficient in dealing with high-dimensional spaces as well as feature extraction because several features can be extracted from one picture.

Suitable For Multi-Class Classification Problems:

SVM is able to perform both binary and multi-class classifications. SVM is capable of detecting different brain tumours including glioma, meningioma, and pituitary tumours.

Processing of MRI-Based Tumour Patterns with Optimal Efficiency:

Using algorithms such as Sequential Minimal Optimization makes SVM capable of handling huge number of features efficiently.

In summary, the possible application of SVM in the classification of brain tumors is that it is capable of identifying the optimal separating line for classification, using the kernel function and analyzing MRI data.

4.2.3 MATLAB Implementation

In MATLAB, the fitcsvm function is used to train SVM models for classification tasks, while fitrsvm is used for regression tasks.

Syntax:

Mdl = fitcsvm (X, Y)

- X: Feature matrix, where each row represents a sample and each column corresponds to a feature.
- Y: Target vector, which contains the class labels for classification or continuous values for regression.
- Mdl: The trained SVM model.

Number of Models: For classification, SVM typically involves training a single model. For multi-class classification, SVM uses strategies like one-vs-one or one-vs-rest to handle multiple classes effectively.

4.5.4 Advantages of Support Vector Machine

- **High Classification Accuracy**: SVM often achieves high accuracy, particularly in situations where classes are well-defined. It is known for its robustness in finding the optimal hyperplane that maximizes the margin, leading to better generalization on unseen data.
- **Effective for Non-Linearly Separable Data**: SVM is powerful for datasets where a linear separation is not possible. By applying the kernel trick, it can project the data into higher-dimensional spaces where a linear decision boundary can be found.
- **Handles High-Dimensional Data**: SVM excels in scenarios where the number of features is much larger than the number of samples. It has proven to be highly effective in fields such as text mining and bioinformatics, where the data is often sparse and high-dimensional.
- **Resistant to Overfitting:** SVM tends to resist overfitting, especially when there is a clear margin of separation between classes. Its design to maximize the margin ensures that the model can generalize well even when the dataset is complex or noisy.
- **Flexibility with Kernels**: The use of different kernel functions gives SVM flexibility in modeling complex relationships. Whether it's a linear, polynomial,

or radial basis function, kernels enable SVM to handle a variety of data types and problems.

- **Robust to Outliers**: Since SVM focuses on the support vectors (the most important data points that define the margin), it is less sensitive to noise and outliers, especially if the margin is maximized.
- **Versatility for Both Classification and Regression:** SVM can be used for both classification (discrete labels) and regression (continuous values), making it adaptable to a wide range of machine learning tasks. It can also be extended for anomaly detection and outlier identification, further expanding its applications.

4.3 K- Nearest Neighbours (KNN)

4.3.1Overview of K-Nearest Neighbours

K-Nearest Neighbours is a popular supervised machine learning algorithm where classification is done according to similarity of examples. KNN algorithm is unlike other algorithms where a model is created first before testing. In this algorithm, training data is stored and then used to find the most similar samples to an input to form predictions. For prediction, the last decision for classification is decided by checking the majority class among those similar data samples.

In this project, the classification of brain MRI images is done using KNN on the data samples obtained after processing the brain images and reducing their dimension. Normalization needs to be done before using this technique because the data features will have varying ranges. This is a suitable algorithm since the MRI images from the same class tend to have similar feature characteristics. K value of 5 is selected as it offers a balance.

4.3.2 Key Features of KNN

• **Based on Similarity**: New instances are classified by comparing them to known examples and selecting the most similar ones.

• **Easier to Comprehend**: Because the basic principle involves the proximity of data, it becomes simple to comprehend.

• **Training Not Required**: As it does not require training because of the storing of data sets rather than learning models, its process is easier.

• **Multi-Class Classification**: It can perform multi-class classifications like tumours such as glioma, meningioma, and pituitary.

• **No Assumption of Data**: It is based on no particular assumption regarding the type of data being used.

4.3.3 MATLAB Implementation

KNN may be applied using MATLAB with the fitcknn function, where fitcknn is used for classification purposes.

Function syntax: Mdl = fitcknn(X,Y)

X: This is the feature matrix, where each row indicates a single observation.

Y: This is the vector containing the class label information.

Mdl: The trained KNN model. In this study, various parameters have been chosen in order to optimize the KNN classifier's performance.

4.3.4 Advantages of KNN

• **Non-Parametric Nature:** KNN does not assume any prior distribution of data, making it suitable for complex and non-linear MRI feature patterns.

• **No Training Phase (Lazy Learning):** The algorithm does not require model training and directly uses stored data for classification, reducing training time.

• **Multi-class Classification Capability:** KNN naturally supports multi-class problems, enabling classification of different tumour types such as glioma, meningioma, and pituitary.

• **Adaptability to Local Data Structure:** Classification is based on nearest neighbors, allowing the model to capture local variations in MRI image features.

• **Flexible Distance Metrics:** KNN supports multiple distance measures such as Euclidean and Manhattan distance, making it adaptable to different feature representations.

• **Effective with Normalized Data:** When combined with normalization techniques, KNN performs efficiently on high-dimensional MRI feature datasets.

4.4 Naïve Bayes

4.4.1 Overview of Naïve Bayes

The Naive Bayes classifier is a classification technique that utilizes probability theory. The technique applies Bayes' theorem, which makes assumptions regarding the independence of attributes to ensure ease of calculations and efficiency. The classification algorithm is very efficient and reliable since it performs classification faster and with high accuracy. It is used for text classification, spam detection, diagnosing diseases, analyzing sentiments, and recommendation systems among others. The Naive Bayes classifier is used in the classification of watermelons based on their images and sounds. Its ability to perform probability computations makes the algorithm fast and efficient enough to classify agricultural produce depending on whether they are ripe or not.

Bayes' theorem mathematically can be represented by:

$$P(C|X) = (P(X|C) * P(C)) / P(X) \quad (4.1)$$

where,

- P(C|X): Posterior probability of class C given feature vector X.
- P(X|C): Likelihood of observing X given class C.
- P(C): Prior probability of class C that can be found even before observing the feature values.
- P(X): Total probability of X over all categories.

The classifier then puts the data points into the class with the highest probability. In the case of Naïve Bayes, sweetness is predicted from the image and acoustic features of watermelons.

4.4.2 Key Features of Naïve Bayes

- **Probabilistic Approach to Classification** It computes the probability of each class, enabling prediction of whether an MRI image belongs to tumour or non-tumour categories based on extracted features such as intensity, texture, and shape patterns.
- **Efficiency**: The algorithm processes multiple features simultaneously and

performs fast classification, making it suitable for handling large MRI datasets with reduced computational time.

- **Effective with Small Training Samples**: Naive Bayes performs well even with limited medical datasets, which is important since labeled MRI data is often scarce and expensive to obtain.
- **Ability to Work with Features of High Dimensionality** MRI images produce high-dimensional feature sets (e.g., pixel intensities, HOG features, PCA-reduced features), and the model efficiently handles such data without complex feature engineering.
- **Multi-class Classification Capability**: It supports multi-class classification, enabling identification of different tumour types such as glioma, meningioma, and pituitary tumours instead of only binary classification.

4.4.3 MATLAB implementation

The naive Bayes algorithm can be implemented using the fitcnb command, which represents the implementation of a naive Bayes classifier for binary and multiclass problems.

Syntax:

Mdl = fitcnb(X, Y)

- X: Input data matrix (features) in which rows represent samples and columns represent features.
- Y: Target data (labels) that could either be binary or categorical.
- Mdl: Trained naive Bayes model.

Number of Models:

Naive Bayes usually has only one model derived from probability computation per label, but if there are more than two labels in the dataset, then multiple distribution models like Gaussian, multinomial, or Bernoulli are applied.

4.4.4 Advantages of Naïve Bayes

• Scalable & Efficient: One great strength of the Naïve Bayes classifier is that it performs efficiently and can easily scale as per the growth in data volume. Thus, this algorithm is ideal for big data applications since it can process data without consuming much computational power.

• High Accuracy with Low Volume Data: As opposed to deep learning algorithms, which require high volumes of labelled data to learn from, the Naïve Bayes algorithm can deliver accurate results using a low number of labelled samples. This makes it useful in domains such as medical studies, where labelled data can be difficult to obtain.

• Robust Against Noise & Missing Features: The ability of the algorithm to handle different features independently renders it resilient against noisy and missing data.

4.5 Summary

The classical machine learning algorithms such as Support Vector Machine (SVM), K Nearest Neighbor (KNN), and Naive Bayes can serve as a stable method for the brain tumor classification problem if they are applied along with the proper data pre-processing and feature extraction methods. SVM is the most effective one among the aforementioned algorithms due to its ability to create distinct decision boundaries as well as work efficiently in high dimensional space. The KNN algorithm functions according to the similarities between the samples, whereas Naive Bayes creates a probabilistic solution.

CHAPTER 5: Deep Learning Models and Soft Voting

5.1 INTRODUCTION

Deep learning techniques are applied in this study in order to circumvent some of the problems associated with conventional machine learning approaches, which rely heavily on hand-engineered features. The ability of Convolutional Neural Networks (CNN) to identify complex patterns like edge detection, texture, tumor detection, among others, makes them very suitable for this task. For this study, a fine-tuned ResNet18 model is adopted.

5.1.1 Convolution Operation

$$(F * X)(i,j) = \sum_{m} \sum_{n} F(m,n) \cdot X(i-m, j-n). \quad (5.1)$$

This operation extracts spatial and textural features such as edges and contours.

5.1.2 Activation (ReLU)

$$f(x) = \max(0, x). \quad (5.2)$$

ReLU introduces non-linearity and accelerates convergence.

5.1.3 Max Pooling

$$Y(i,j) = \max_{(m,n)\in \text{window}} X(i+m, j+n). \quad (5.3)$$

Pooling reduces dimensionality while preserving dominant features.

5.1.4 ResNet18 and Residual Learning

Residual learning introduces skip connections defined as:

$$y = F(x, W_i) + x. \quad (5.4)$$

The gradient flows through:

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial y}\left(1 + \frac{\partial F}{\partial x}\right). \quad (5.5)$$

This improves training stability and prevents gradient vanishing, enabling deeper CNN architectures like ResNet18.

5.1.5 Fine-Tuning

Let θ_{pre} be pretrained weights and θ_{new} the updated weights after fine-tuning:

$$\theta \leftarrow \theta - \eta \cdot \nabla_{\theta} L. \quad (5.6)$$

Frozen layers retain:

$$\theta_{\text{frozen}} = \theta_{\text{pre}}. \quad (5.7)$$

5.2. Custom CNN (Fine tuned Resnet-18)

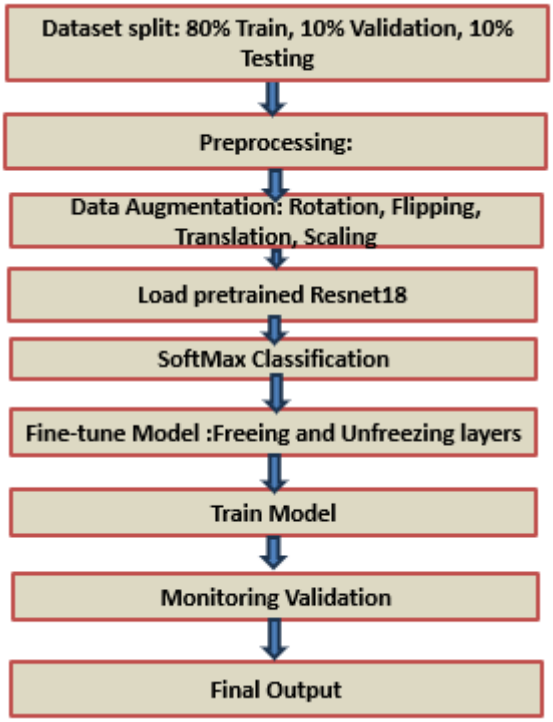


Fig 5.2 Block diagram for Fine-tuned Resnet-18

5.2.1 Deep learning Pipeline

• Dataset splitting

The split ratio of the dataset includes 80%, which will be used for training; 10% for validating, and 10% for testing. This process is aimed at training and validating the model to check its performance.

• Preprocessing

Before the image is fed into the model, it is preprocessed by resizing it to a standard size of 224 × 224 × 3.

• Data augmentation

Data augmentation techniques such as rotating, flipping, translating, and scaling the training dataset are carried out to improve the diversity of the dataset.

• Load pretrained ResNet18

As mentioned above, a pretrained ResNet18 will be used because the network will have already learned valuable low-level features from a big amount of data, such as edges and textures.

• Softmax classification

The output from the model will contain an unnormalized log-probability of a sample. This output layer is then modified into a softmax classification to obtain the probability of a particular tumour class.

• Fine-Tuning (Freezing and Unfreezing Layers)

For adaptation to the MRI images dataset, the previous layers are frozen, and the later layers are slightly unfrozen. That way, the model is allowed to use learned features while developing features specific to tumour identification.

• Train Model

In training the network, we use Adam with a very low learning rate for the optimization process. The training will be performed by tuning the network's weight according to the provided training samples.

• Monitor Validation

While training, the accuracy of the validation set is regularly checked. Monitoring the accuracy of validation enables us to determine whether our model is overfitting.

• Final Output

When the training process is done, we evaluate the model on the testing data set. In the end, we get the predicted classes with the accuracy of prediction among others.

5.2.2 Training Configuration

The network is trained based on the following specifications:

- Optimizer: Adam
- Learning Rate: 3×10^{-5}
- Epochs: 25
- Batch Size: 32
- Learning Rate Drop: 0.5
- Drop Frequency: Every 3 epochs

These specifications make the training process stable and effective.

5.2.3 Early Stopping

A stopping criterion is employed to avoid overfitting. In case there is no improvement in the validation accuracy after a set number of iterations, the training process is automatically halted. This saves computational efforts and enhances generalizability.

5.2.4 Model Evaluation

Testing the performance of the trained model is done using test data. Accuracy serves as a metric for performance measurement, while the confusion matrix facilitates analysis based on classes.

Table 5.2.4**:** Architecture details of fine-tuned Resnet-18

Stage	Layer Type	Filter Size/Parameters	Output Size	Description
Input	-	-	224x224x3	Input MRI Brain Image
Conv1	Convolution + BN + ReLU	7x7,64Filters, Stride 2	112x112x64	Initial feature extraction
MaxPool	Max Pooling	3x3, stride 2	56x56x64	Down samples spatial dimensions
Conv2_x-Block 1	2xConv+BN+ReLU+Skip	3x3, 64 filters	56x56x64	Residual Block 1
Conv2_x-Block 2	2xConv+BN+ReLU+Skip	3x3, 64 filters	56x56x64	Residual Block 2
Conv3_x-Block 1	2xConv+BN+ReLU+Skip	3x3, 128 filters	28x28x128	Residual Block 3
Conv3_x-Block 2	2xConv+BN+ReLU+Skip	3x3, 128 filters	28x28x128	Residual Block 4
Conv4_x-Block 1	2xConv+BN+ReLU+Skip	3x3, 256 filters	14x14x256	Residual Block 5
Conv4_x-Block 2	2xConv+BN+ReLU+Skip	3x3, 256 filters	14x14x256	Residual Block 6
Conv5_x-Block 1	2xConv+BN+ReLU+Skip	3x3, 512 filters	7x7x512	Residual Block 7
Conv5_x-Block 2	2xConv+BN+ReLU+Skip	3x3, 512 filters	7x7x512	Residual Block 8
AvgPool	Global Average Pooling	-	1x1x512	Averages each feature map
FC	Fully Connected	512-2 neurons	1x1x2	Tumour no tumour classification
Softmax	Softmax Layer	-	1x1x2	Probability distribution output
Output	Classification Layer	-	-	Final predicted class label

5.3 Soft Voting Ensemble for Binary Classification

5.3.1 INTRODUCTION

The initial step is data pre-processing. The second step is augmentation of the data to promote improved learning. Training of multiple CNN models is done at the third stage. SV algorithm is used in model selection and fusion. An intelligent approach to brain tumour detection is proposed

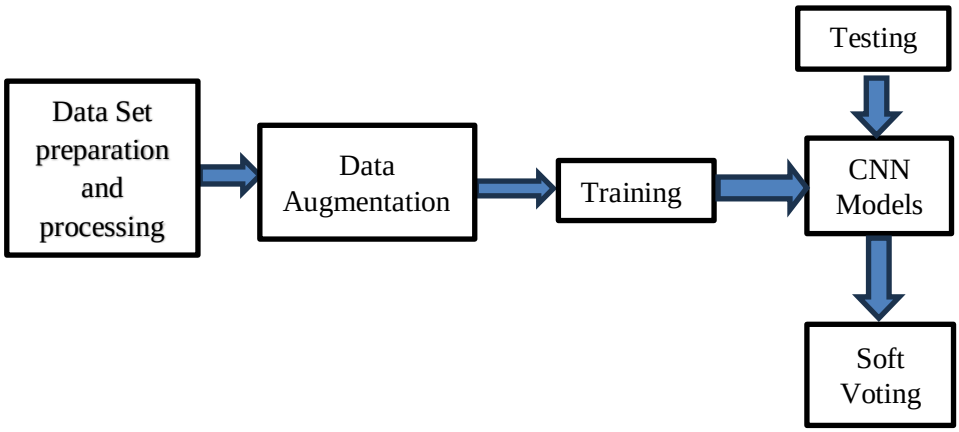


Fig 5.3. Block diagram for soft voting

5.3.2 Dataset Preparation and Preprocessing

MRI image sets can be classified into two types in this study [11]. These are Tumour and No Tumour. Proper labelling was done by an expert panel of doctors. The dataset can be categorised into two sets. They are training and testing sets. The training dataset uses 70% of the dataset. Testing uses the remaining 30%. The images are scaled to 224 x 224.

5.3.3 Data Augmentation

This diversity is enhanced through data augmentation techniques. Some of the data augmentation techniques include rotation, flipping, scaling, and resizing. The new data set will allow for learning of new features for classification purposes. This

helps the CNN model detect important patterns within the data set. This will also help mitigate the problem of overfitting.

5.3.4 Architecture of Proposed Soft Voting Ensemble Model

The differences in brightness, lighting, or sharpness are observed in medical images. The pre-processing helps enhance image quality. The proposed network enhances the classification process. The most important features are selected to enhance the accuracy rate. Each model has its unique ability to detect specific patterns. The experiments include the use of five different models, including MobileNetV2, ResNet18, ResNet50, DenseNet201, and EfficientNetB0. All models are capable of classifying two categories. They include C1(Tumour) and C2(No Tumour). Proposed architecture adopted in this work is shown in Fig.2.

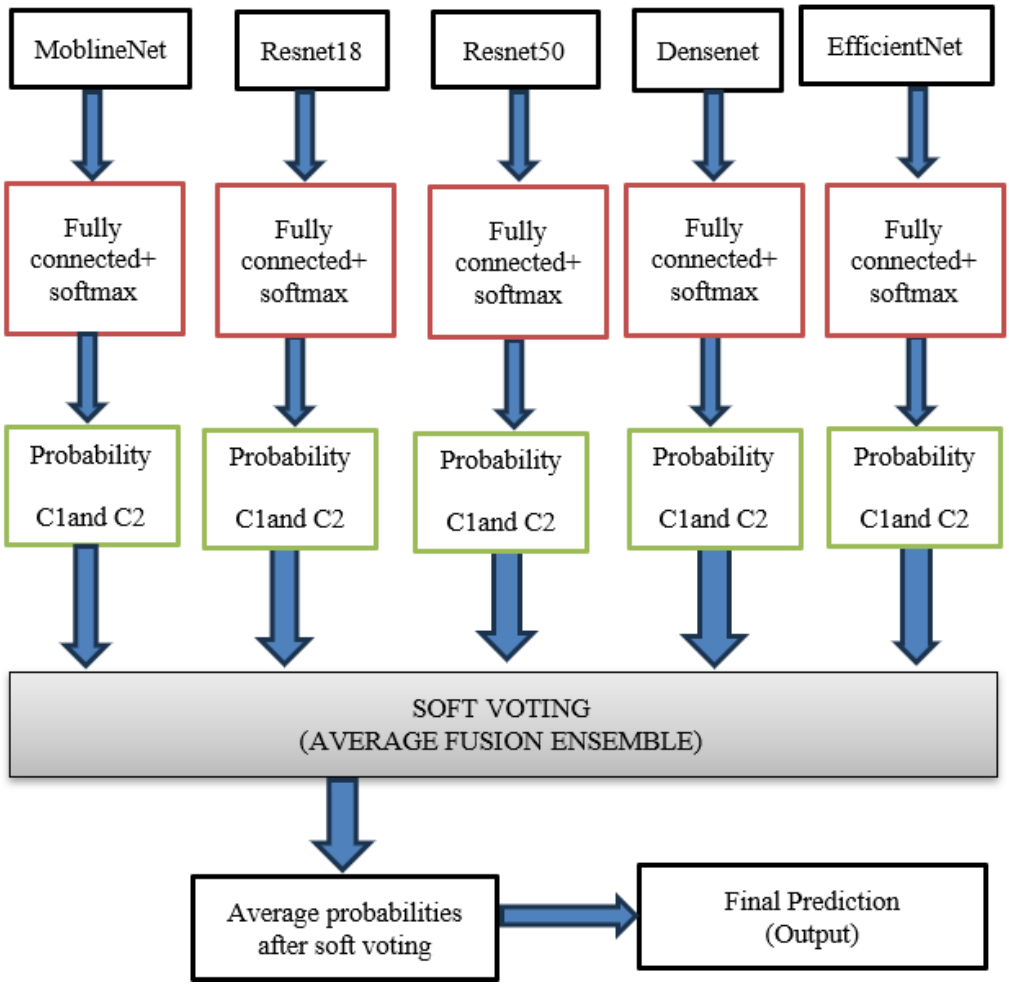


Fig 5.4 Proposed Architecture for soft voting in view of binary classification

The training and testing processes followed by all the models are the same. The depth of each of the models can be varied. The early layers capture basic information. With moving towards the final layers, the model starts identifying global features. Various types of filters are used in this process. These filters help to capture various types of patterns. ReLU activation function is applied. Layers called pooling layers help to keep the size under control. The convolutional layers perform the job of identifying the features one by one at each stage in every CNN model. Every CNN model uses a similar procedure. Though there is a difference in the depth of the architecture, the procedure of the operation is common. In the early stages, the model learns basics such as lines, change in intensity levels or contrasts which are important for detecting areas in the brain that come together. Soft max is used for the final classification. Averaging probability and then predicting is helpful to improve accuracy so that it shows how effective the training process has been. Batch normalization along with early stopping helps prevent memorization and learning excessively. This helps the model to remain stable and minimize training oscillations.

Each CNN model within the ensemble has unique design characteristics:

MobileNetV2

The MobileNetV2 model represents a lightweight architecture for deep CNNs focused on performance and computational efficiency. Depth wise separable convolution layers and inverted residual structures are employed to minimise computation without sacrificing performance accuracy. The model successfully captures fine details and performs optimally on low-contrast MRI scans.

ResNet18 and ResNet50

The ResNet model uses residual skip connection. The use of such skip connection helps in passing the gradient along the identity mapping, helping to solve the vanishing gradient issue.

• ResNet18 consists of 18 convolutional layers. It is designed for learning general features. It works well for small data sets.

• ResNet50 is more complex compared to ResNet18. It is used in identifying complicated features within large data sets.

DenseNet201

The strong relationship between the layers is employed. The use of a feature reuse concept enhances the performance of the network. The hierarchical nature of the features allows the difference in intensity to be detected.

EfficientNetB0

Scaling technique is used in this model. Depth, width, and resolution of the network are adjusted together

5.3.5 Network Flow

Description of the architecture is briefly provided below. Input: MRI image(224×224×3). Convolutional Layers:Feature extraction on spatial data with different kernels(3×3,5×5). Batch Normalization and ReLU Activation:Activation normalization and nonlinearity addition. Pooling Layers:Spatial reduction with retaining significant features. Fully Connected Layers:Feature combination and classification task. Softmax Layer:Probability generation from raw scores. Soft Voting Fusion:Fusion of probability values from all five CNNs to make a final decision. The core contribution of this research can be considered the Soft Voting ensemble learning approach that fuses probabilities from all five trained CNN models to make a final classification decision.

5.3.6 Mathematical Representation

Let the dataset be represented as

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{M} \tag{5.8}$$

where $x_i \in \mathbb{R}^{224\times224\times3}$is the i^{th}MRI image and $y_i \in \{0,1\}$is the corresponding label, with

- $y_i = 1$representing *Tumour*, and
- $y_i = 0$representing *No Tumour*.

The ensemble consists of $N = 5$pretrained CNN models:

$$\mathcal{M} = \{M_1, M_2, M_3, M_4, M_5\}$$

where each $M_k$corresponds to one of the models — MobileNetV2, ResNet18, ResNet50, DenseNet201, and EfficientNetB0. Each model $M_k$extracts hierarchical feature maps $F_k$from the input image $x_i$through multiple convolutional operations:

$$F_k = f_k(x_i; \theta_k) \tag{5.9}$$

where θ_k represents the learnable parameters (weights and biases) of model M_k. The output of the final fully connected layer for model M_k is a 2-dimensional logit vector:

$$Z_k = [z_{k,0}, z_{k,1}] \quad (5.10)$$

where $z_{k,0}$ and $z_{k,1}$ are the raw scores for *No Tumour* and *Tumour*, respectively. Each model's logits are passed through a Softmax layer to generate normalized probability values:

$$P_{k,j} = \frac{e^{z_{k,j}}}{\sum_{c=1}^{C} e^{z_{k,c}}}, j \in \{0,1\} \quad (5.11)$$

where:

- $P_{k,0}$ = probability of *No Tumour* predicted by model M_k,
- $P_{k,1}$ = probability of *Tumour* predicted by model M_k,
- $C = 2$ = number of classes.

For each test image, each model produces a probability vector:

$$P_k = [P_{k,0}, P_{k,1}] \quad (5.12)$$

The ensemble model fuses the probabilistic outputs of all CNNs using arithmetic mean fusion, also known as Soft Voting:

$$\hat{P}_j = \frac{1}{N} \sum_{k=1}^{N} P_{k,j}, j \in \{0,1\} \quad (5.13)$$

Thus, for the two-class case:

$$P\hat{}_0 = 1/5\ (P_1,0 + P_2,0 + P_3,0 + P_4,0 + P_5,0\ ) \quad (5.14)$$

$$\hat{P}_1 = \frac{1}{5}\left(P_{1,1} + P_{2,1} + P_{3,1} + P_{4,1} + P_{5,1}\right) \quad (5.15)$$

where:

- $\hat{P}_0$ = ensemble probability of *No Tumour*,
- $\hat{P}_1$ = ensemble probability of *Tumour*.

The final class prediction for image x_i is determined by selecting the class with the highest ensemble probability:

$$\hat{y}_i = \arg\max_{j \in \{0,1\}} \left(\hat{P}_j\right) \quad (5.16)$$

Hence, $\hat{y}_i = \begin{cases} 0, & \text{if } \hat{P}_0 > \hat{P}_1 \\ 1, & \text{otherwise} \end{cases}$

5.4 MOBILENET V2

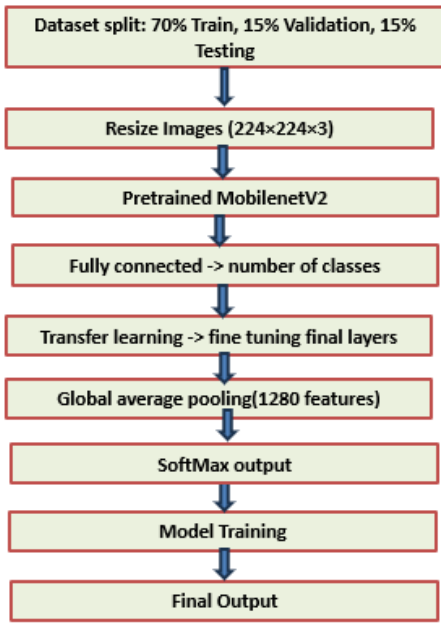


Fig 5.4 Block diagram of MOBILENET V2

5.4.1 Deep Learning Pipeline (MobileNetV2)

• Dataset Splitting

This split of the data set into three parts helps to train the model properly as well as test the model on unseen data to find out how well the model has generalised the information.

•Preprocessing

Prior to training on the images using the model, all the images from the MRI scans are standardised to have the same input size that is necessary for MobileNetV2.

• Load Pretrained MobilenetV2

MobileNetV2 has been pre-trained to act as the backbone network. As it has been trained on large-scale datasets, it is capable of extracting useful low-level features like edges and textures, making it suitable for brain tumour recognition.

• Modify Final Layers

These two layers have been substituted with other layers as per the number of tumour categories. In these layers, high learning factor rates are set so that these layers get adapted quickly to the new dataset.

• Softmax Classification

Finally, the last layers of the neural network are adjusted by incorporating a softmax layer to generate probabilities for each tumour category. The category with the maximum probability will be considered as the final prediction.

• Train Model

The model is trained with the help of the Adam optimiser with GPU assistance to speed up the computations. The batch size is set at 64, while the maximum number of epochs is limited to 8, which facilitates fast and effective training.

• Monitoring Validation

During training, validation accuracy is monitored at regular intervals. This helps track the model's performance and ensures that it is not overfitting.

• Final Output

After training is complete, the model is evaluated on test data. The final output includes predicted tumour classes along with performance metrics such as accuracy.

5.4.2 Training Configuration

The model is trained using the following parameters:

• Optimiser: Adam

• Execution Environment: GPU

• Learning Rate: 5×10^{-4}

• Epochs: 8

• Batch Size: 64

• Shuffle: Every epoch

• Validation Data: Used during training

• Validation Patience: 2

These settings are selected to ensure fast convergence and efficient training while maintaining stable performance.

5.4.3 Early Stopping

Early stopping is used as a technique to avoid overfitting. If the accuracy on the validation set does not increase after a certain number of steps, then the training will

be automatically stopped.

5.4.4 Model Evaluation

The trained model is evaluated using test data. Accuracy is calculated to measure overall performance, and a confusion matrix is used to analyse class-wise predictions

Table 5.4.4: Architecture details of Mobile Net V2

Stage	Block	Kernel	Expansion(t)	Filters(c)	Repeats(n)	Stride(s)	Output Size
Input	-	-	-	-	-	-	224x224x3
Stem	conv	3x3	-	32	1	2	112x112x32
1	Bottle Neck	3x3	1	16	1	2	112x112x32
2	Bottle Neck	3x3	6	24	2	2	56x56x24
3	Bottle Neck	3x3	6	32	3	2	56x56x24
4	Bottle Neck	3x3	6	64	3	2	28x28x32
5	Bottle Neck	3x3	6	96	4	2	14x14x64
6	Bottle Neck	3x3	6	160	3	2	14x14x96
7	Bottle Neck	3x3	6	320	3	2	7x7x160
Head	Conv	1x1	-	1280	1	1	7x7x1280
Pool	Global Avg Pool	-	-	-	-	-	1x1x1280
PC		-	-	3	-	-	3 Classes

5.5 Custom EFFICIENTNET-B0

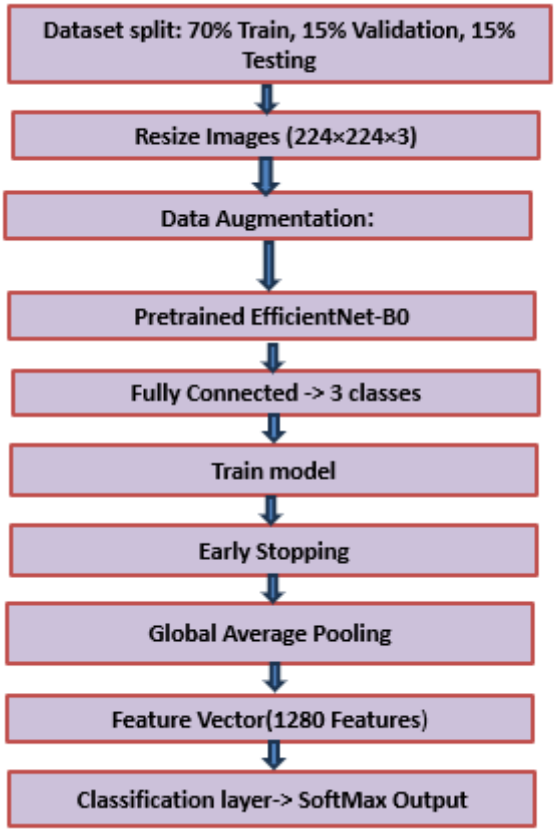


Fig 5.5 Block Diagram for Custom Efficient Net B0

5.5.1 Deep Learning Pipeline(Efficient net B0)

• Dataset Splitting

This data set is read through the use of MATLAB's image datastore function, where automatic labeling takes place based on the names of the folders. It is initially split into 70% for training purposes, while the other 30% is evenly split between the validation and testing datasets at 15% each.

•Preprocessing

All MRI images are resized according to the input size required by EfficientNet-B0 using augmented Image Datastore. This ensures consistency across the dataset and allows smooth processing within the network.

• Data Augmentation

Data augmentation is performed only on the training set through the use of image DataAugmenter. The transformation consists of random rotation of −15° to +15°, random horizontal flip, random scaling of 0.9 to 1.1, and random translation by ±10 pixels in X and Y axes.

• Load Pretrained EfficientNet-B0

The EfficientNet-B0 model is loaded as a pretrained network. It is designed to efficiently scale depth, width, and resolution, allowing it to extract both simple and complex features from MRI images.

• Modify Final Layers

Replacement of the original fully connected layer and classification layer is done through the use of replaceLayer function. The new fully connected layer is generated, where its output size will be equivalent to the number of tumor classes, along with higher learning rate factors to ensure fast adaptation.

• Softmax Classification

The last two layers of the neural network are altered such that a softmax layer is added to make the output into probabilities for each class of tumor. The class having the maximum probability will be considered the output class.

• Train Model

The training process for the neural network employs the trainNetwork method using the Adam optimization algorithm. The maximum number of training epochs is set to 50, with a batch size of 32 during each iteration. An initial learning rate of 1×10^{-4} guarantees stable learning while facilitating convergence.

• Monitoring Validation

During training, validation accuracy is monitored at regular intervals. This helps track the model's performance and ensures that it is not overfitting.

• Final Output

After training is complete, the model is evaluated on test data. The final output includes predicted tumour classes along with performance metrics such as accuracy.

5.5.2 Training Configuration

The model is trained using the following parameters:

• Optimizer: Adam

• Execution Environment: Auto

• Learning Rate: 1×10^{-4}

• Epochs: 50

• Batch Size: 32

• Validation Frequency: 30

• Validation Patience: 7

• Learning Rate Schedule: Piecewise

• Learning Rate Drop Factor: 0.5

• Drop Period: Every 10 epochs

• L2 Regularization: 0.0005

• Shuffle: Every epoch

These settings are selected to ensure fast convergence and efficient training while maintaining stable performance.

5.5.3 Early Stopping

An early stopping mechanism is implemented using the validation patience parameter. If the validation accuracy does not improve for 7 consecutive checks, the training process is stopped automatically. This helps prevent overfitting and reduces unnecessary computation.

5.5.4 Model Evaluation

The trained model is evaluated using test data. Accuracy is calculated to measure overall performance, and a confusion matrix is used to analyse class-wise predictions.

Table 5.5.4: Architecture Details of Custom Efficient Net B0

Stage	Block	Kernel	Filters	Stride	Output size
Input	-	-	-	-	224x224x3
Stem	conv	3x3	32	2	112x112x32
1	MBconv1	3x3	16	1	112x112x16
2	MBconv6	3x3	24	2	56x56x24
3	MBconv6	5x5	40	2	28x28x40
4	MBconv6	3x3	80	2	14x14x80
5	MBconv6	5x5	112	1	14x14x112
6	MBconv6	5x5	192	2	7x7x192
7	MBconv6	3x3	320	1	7x7x320
Head	Conv	1x1	1280	1	7x7x1280
Pool	GAP	-	-	-	1x1x1280
FC	Dense	-	3	-	3 classes

5.6 Custom Mobile Net

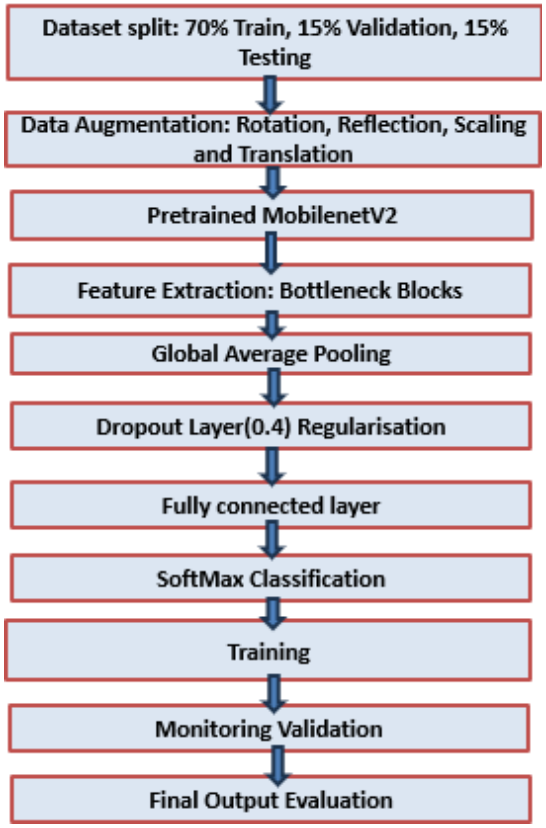


Fig 5.6 Block Diagram for Custom Mobile net

5.6.1 Deep Learning Pipeline

• Dataset Splitting

The dataset is loaded using MATLAB's imageDatastore, where images are labeled automatically based on folder names. It is divided into 70% training data, and the remaining 30% is split equally into validation and testing sets (15% each). This setup ensures effective training while also allowing proper evaluation on unseen data.

•Preprocessing

All MRI images are resized to match the input size required by MobileNetV2 using augmentedImageDatastore. This ensures uniformity across the dataset and allows the network to process images efficiently.

• Data Augmentation

Data augmentation is applied to the training dataset using imageDataAugmenter. The transformations include random rotation (−15° to +15°), horizontal reflection, scaling (0.9 to 1.1), and translation in both directions within ±8 pixels. These variations help improve model robustness and reduce overfitting.

• Load Pretrained MobilenetV2

A pretrained MobileNetV2 model is used as the base network. It is capable of extracting important visual features such as edges, textures, and patterns, which are useful for brain tumour classification.

• Modify Network Architecture

The original classification layers are modified by replacing the fully connected layer (Logits) with a new layer corresponding to the number of tumour classes. A new classification layer is also introduced. In addition, a dropout layer with a rate of 0.4 is inserted after the global average pooling layer to reduce overfitting. The network connections are updated accordingly using disconnectLayers and connectLayers.

• Softmax Classification

The final layers of the network are modified to include a softmax layer, which converts the output into probabilities for each tumour class. The class with the highest probability is selected as the final prediction.

• Train Model

The model is trained using the trainNetwork function with the Adam optimizer and GPU acceleration. Training is performed for 35 epochs with a mini-batch size of 32. An initial learning rate of $1 \times 10^{-4}$is used, along with L2 regularization to improve generalization.

• Monitoring Validation

During training, validation accuracy is monitored at regular intervals. This helps track the model's performance and ensures that it is not overfitting.

• Final Output

After training is complete, the model is evaluated on test data. The final output includes predicted tumour classes along with performance metrics such as accuracy.

5.6.2 Training Configuration

The model is trained using the following parameters:

• Optimizer: Adam

• Execution Environment: GPU

• Learning Rate: 1×10^{-4}

• Epochs: 35

• Batch Size: 32

• Validation Patience: 6

• Learning Rate Schedule: Piecewise

• Learning Rate Drop Factor: 0.4

• Drop Period: Every 10 epochs

• L2 Regularization: 0.0005

• Shuffle: Every epoch

These settings are selected to ensure fast convergence and efficient training while maintaining stable performance.

5.6.3 Early Stopping

An early stopping mechanism is implemented using validation patience. If the validation performance does not improve for 6 consecutive checks, the training process is stopped automatically. This helps prevent overfitting and reduces unnecessary computation..

5.6.4 Model Evaluation

The trained model is evaluated using test data. Accuracy is calculated to measure overall performance, and a confusion matrix is used to analyse class-wise predictions

Table 5.6.4**:** Architecture Details of Custom Mobile Net

	Stage	Block	Kernel	t factor	#Filters(c)	Repeats(n)	Outputsize
Input	-	-	-	-	-	-	224x224x3
Stem	Conv	Conv	3x3	-	32	1	112x112x32
S1	S1	MBConv1	3x3	1	16	1	112x112x16
S2	S2	MBConv6	3x3	6	24	2	56x56x24
S3	S3	MBConv6	3x3	6	32	3	28x28x32

S4	S4	MBConv6	3x3	6	64	4	14x14x64
S5	S5	MBConv6	3x3	6	96	3	14x14x196
S6	S6	MBConv6	3x3	6	160	3	7x7x160
S7	S7	MBConv6	3x3	6	320	1	7x7x320
Head	Conv	1x1	-	-	1280	1	7x7x1280
Pool	GAP	-	-	-	-	1	1x1x1280
Dropout	Dropout (0.4)		-	-	-	3	1x1x1280
FC	Dense		-	-	3	-	3 classes

5.7 Comparison of Deep Learning Models

Comparison of the three deep learning models will be done on the basis of architectural design, training settings, and generalization capabilities. All three models use the same image dimensions (224 x 224 x 3) and utilize the Adam optimizer for training, but they perform differently because of the difference in regularizers, data augmentation techniques, and learning approaches.

Training this model takes a shorter time and includes a simpler training procedure that has low complexity, which implies the use of few epochs without dropout or augmentation. Although such an approach facilitates fast convergence, this model lacks generalization capabilities due to the absence of dropout and weak augmentation and provides worse results than those of Custom MobileNet and EfficientNet-B0.

Compared to MobileNetV2, the Custom MobileNet architecture adds a dropout layer (0.4) after the global average pooling layer and implements more significant L2 regularisation (5×10^{-4}). Moreover, additional data augmentation is used to rotate images (±15°), scale them (0.9–1.1) and translate (±8 pixels). Such changes enable faster but more stable convergence during training on 35 epochs with the piecewise learning rate schedule (factor drop – 0.3 every 10 epochs). Thus, overfitting is reduced significantly, and feature learning becomes more effective. As a result, 95.65% accuracy is achieved in classification.

EfficientNet-B0 utilises compound scaling when scaling up network depth, width, and input resolution systematically. For training, this model uses a larger learning rate (1×10^{-4}) with a batch.

Table 5.7**:** Training operations and specifications Comparison

Parameter	Mobile Net V2 (Pretrained)	Custom Mobile Net (Modified)	Efficient Net B0 (Pretrained)
Batch Size	32	32	32
Initial Learning Rate	1e-4	1e-4	1e-4
Optimizer	Adam	Adam	AdamW
L2 Regularization	1e-4	5e-4	1e-5
Dropout	None	0.4	0.3
Data Augmentation	-	Advanced (Rotation, Flip, Scale, Translation)	Basic (flip, rotation)
Learning Rate Schedule	Step Decay (Drop=0.3, Every 7 Epochs)	Step Decay (Drop=0.3, Every 7 Epochs)	Cosine Decay
Early stopping patience	Epochs	6 Epochs	5 Epochs
Shuffle	Every Epoch	Every Epoch	Every Epoch
Execution Environment	GPU	GPU	GPU
Input Size	224x224x3	224x224x3	224x224x3

5.8 Soft Voting for Multiclass Classification

In order to improve the results of classification, a hybrid soft voting technique is used by aggregating the predictions made by the EfficientNet-B0 and Custom MobileNet models. Initially, the two trained models are loaded and tested against the validation set in order to obtain the performance level of each model. The validation accuracy of each model is used to assign adaptive weights to the models, making sure that the more accurate model will have more weight in the decision-making process.

For testing purposes, the probabilities for each class are obtained through the predict method for each model. However, instead of averaging the probabilities, a hybrid approach is used to make a decision. For each sample, the maximum probability (or confidence) is calculated between the two models. If one of the models gives a very high confidence (>0.85), then the prediction is accepted as it is.

If none of the two models has high confidence levels, a weighted soft voting algorithm is applied, whereby the output probabilities are averaged based on weight coefficients. The weight coefficients are calculated based on validation accuracy,

which ensures that at any point in time, the more accurate model gets priority over the other one. The resulting class prediction will be based on the class that receives the highest average probability.

5.9 Summary

A combination of hybrid soft voting is adopted in which predictions made by EfficientNet-B0 and Custom MobileNet models are taken into account. First, accuracies of both models are estimated based on validation results. Accordingly, adaptive weights are computed, providing greater weight to higher accuracy model. In the process of testing, models provide predictions of probability for all classes. A confidence level is employed to determine the most accurate class. If any model makes a prediction with very high confidence levels (probability over 0.85), the prediction is considered true. If no such high-confidence probability is detected, then weighted soft voting method is applied. According to this approach, probability from two models is summed up considering adaptive weights and a class having a maximal probability is selected. This method provides high confidence and efficiency of predictions at the same time

CHAPTER 6: RESULTS

The performances of the suggested brain tumour classifier are assessed in two stages using classical machine learning models and deep learning models before applying ensemble techniques for improving performance.

At first, some classical machine learning models like SVM, KNN, and Naive Bayes have been utilized on features obtained from preprocessing steps such as histogram equalization, segmentation, and PCA. It has been found out that SVM produces the most effective results compared to the other two classifiers for binary classification where SVM attains 96.95% accuracy, while KNN and Naïve Bayes attain 85.79% and 59.14%, respectively.

As the use of deep learning techniques for tumour detection is adopted, a remarkable improvement in accuracy is achieved. Fine-tuning the ResNet18 model gives an accuracy rate of 98.73% with both precision (100%) and specificity (100%) rates being one hundred percent. Therefore, the network shows its ability to learn intricate spatial structures from MRI images.

With regard to multiclass classification, there is a notable difference in performances between classical machine learning techniques and deep learning models. Classical machine learning algorithms like SVM produce low accuracies of 75.43%, while KNN and Naïve Bayes even have lesser accuracies at 67.47% and 65.74%, respectively. However, deep learning models exhibit outstanding performances of 91.55%, 94.20%, and 95.65% for MobileNetV2, Custom EfficientNet-B0, and Custom MobileNet, respectively.

Further improvements in the performance of these models are possible through soft voting ensembles where various deep learning models are combined. In binary classification cases, the use of soft voting ensemble gives the best performance metrics, with accuracy at 99.49%, along with precision of 99.55% and recall of 99.45%.

However, when it comes to multiclass classification, even though ensemble methods enhance model efficiency, the enhancement is relatively modest because it requires differentiating multiple kinds of tumours. This implies that model

consensus is naturally higher in binary classification, resulting in superior ensemble performance.

Moreover, a mixed soft voting technique is proposed through EfficientNet-B0 and Custom MobileNet based on confidence rules and adaptive weights. In this regard, model accuracy is further improved since the predictions with the highest confidence level are preferred, and weighted averages are applied in uncertain conditions.Overall, it can be concluded that there is a clear progression in the performance gain from classical machine learning to deep learning and then to ensemble techniques. SVM achieves the highest efficiency in comparison with other ML models, while deep learning models show a remarkable superiority compared to other methods, especially in multiclass scenarios. The optimal efficiency is achieved in the implementation of ensemble techniques, especially in binary classification problems.

6.1 Performance Metrics

6.1.1 Confusion Matrix

A confusion matrix serves to give an overall account of the effectiveness of the model through comparison between the real classes and predicted classes. It also helps in analyzing the strong and weak aspects of the model through calculation of different performance measures such as accuracy, precision, recall, and F1 score.

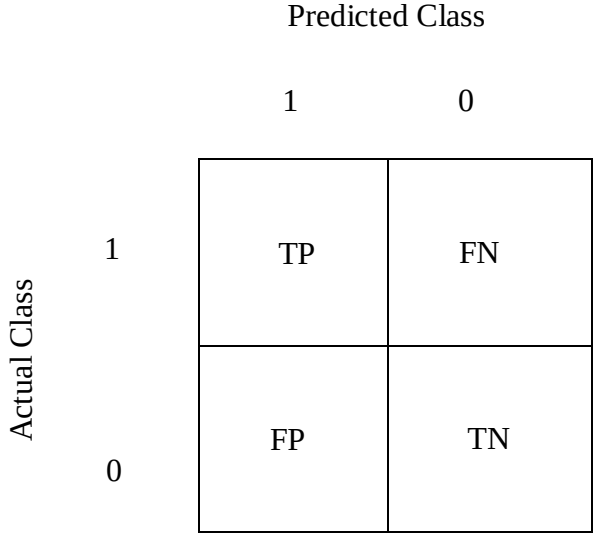


Fig 6.1 Confusion Matrix

- True Positive (TP):

  When the image is actually “Tumor”, but the model predicts the image as "Tumor".

- False Positive (FP):

  When the image is actually “No Tumor”, but the model predicts the image as "Tumor".

- True Negative (TN):

  When the image is actually “No Tumor”, but the model predicts the image as "No Tumor".

- False Negative (FN):

  When the image is actually “Tumor”, but the model predicts the image as "No Tumor".

6.1.2 Accuracy

Accuracy is a basic measurement criterion that reflects the ratio of correctly categorized cases among all the cases. It serves as a general criterion for the correctness of the algorithm and is extensively applied in measuring the performance of algorithms.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (6.1)$$

6.1.3 Precision

Precision, alternatively referred to as Positive Predictive Value, is a measure that evaluates the number of positive results among all positive predictions made. It mainly concentrates on reducing the number of false positives and is very relevant in cases with costly false positives.

$$Precision = \frac{TP}{TP+FP} \quad (6.2)$$

6.1.4 Recall

"Recall" is another name for sensitivity or the True Positive Rate, which calculates how well the model predicts the presence of all the important positives. Recall is

concerned with reducing the number of false negatives, which become significant in critical scenarios.

$$Recall = \frac{TP}{TP+FN} \quad (6.3)$$

6.1.5 F1 Score

F1 Score is the harmonic mean of precision and recall, providing a balance between the two measures by taking into account both false positives and false negatives. The use of F1 Score becomes very important whenever there is a requirement to strike a balance between the two.

$$F1\ Score = \frac{2*Precision*Recall}{Precision+Recall} \quad (6.4)$$

6.2 Binary Classification

6.2.1 SVM

Table 6.2.1. Confusion matrix of SVM

	Predicted no tumor	Predicted tumor
Actual no tumor	93	12
Actual tumor	0	289

From the confusion matrix above, it is clear that the model is quite effective in classifying images of brain tumours. Out of the total number of observations, 93 non-tumour images were accurately identified, while 12 were wrongly categorized as tumours. On the other hand, all 289 images with tumour were accurately predicted, with zero images being falsely predicted as no tumour,

In view of the above, the overall accuracy rate of the model stands at 96.95%, which means that the model performs effectively in terms of classification. With regard to precision for tumors, it is estimated at 96.01%, which shows that the majority of images predicted as tumors are accurate. The recall (or sensitivity) is 100%, which

means that all the actual tumor images are accurately predicted by the model. In terms of specificity, it is at 88.57%.

6.2.2 KNN

Table 6.2.2. Confusion matrix of KNN

	Predicted no tumor	Predicted tumor
Actual no tumor	93	12
Actual tumor	0	289

The confusion matrix shows that the KNN model works well for brain tumour binary classification, especially for detecting the presence of a brain tumour. Specifically, there are 93 samples of non-brain tumours correctly classified, while 12 of them are wrongly classified as having brain tumours. Moreover, there are 289 samples of brain tumours correctly classified, with no error in detecting the presence of a brain tumour in the dataset.

According to the confusion matrix, the accuracy of the model is estimated at 96.95%, which is relatively high. Additionally, the precision of brain tumour detection can be estimated at 96.01%. The recall of brain tumour detection is 100%, which is essential since there is no false negative case in the dataset. The specificity of the model is 88.57%, which implies that only a few samples of non-brain tumours were falsely classified as brain tumours.

6.2.3 Naïve Bayes

From the confusion matrix, it can be observed that the Naïve Bayes algorithm gives excellent classification for binary brain tumours in terms of classification as a tumour. There are 93 cases where there is no tumour which are correctly identified while there are only 12 instances where a tumour is falsely predicted. On the other hand, all 289 cases where there is a tumour have been correctly classified with no errors or false negatives.

Table 6.1.3. Confusion matrix of Naïve Bayes

	Predicted no tumor	Predicted tumor
Actual no tumor	93	12
Actual tumor	0	289

The overall accuracy of the algorithm has been found to be approximately 96.95% which means that there has been excellent classification achieved through this model. As far as the tumour precision is concerned, it has been calculated at approximately 96.01%. The recall rate has been calculated to be 100%. This is very important in terms of medical application as there should be no error on this side. Specificity has been calculated at approximately 88.57%. The F1-score of approximately 97.97% has also been found for the model.

6.2.4 Custom CNN(Fine-tuned Resnet-18)

From the confusion matrix above, we can conclude that the Custom CNN (fine-tuned ResNet18) model achieves an extremely high level of accuracy in the detection of brain tumours. Out of 110 samples that were not affected by the disease, 105 are correctly identified as non-tumour, while 5 are incorrectly classified as tumours. On the other hand, the model detects 284 out of 284 instances as tumours, implying a perfect accuracy rate in terms of identification. From the confusion matrix above, we can conclude that the Custom CNN (fine-tuned ResNet18) model achieves an extremely high level of accuracy in the detection of brain tumours. Out of 110 samples that were not affected by the disease, 105 are correctly identified as non-tumour, while 5 are incorrectly classified as tumours. On the other hand, the model detects 284 out of 284 instances as tumours, implying a perfect accuracy rate in terms of identification.

Table 6.2.4. Confusion matrix of fine-tuned Resnet18

	Predicted no tumor	Predicted tumor
Actual no tumor	105	5
Actual tumor	0	284

With such accuracy rates, the overall model accuracy stands at about 98.73%, which makes it a reliable classifier. Tumour detection precision is approximately 98.27%, suggesting that nearly all tumours identified as positive samples by the model are actually affected. Recall (or sensitivity) is 100%, meaning that all tumours were accurately detected**.**

6.3 Multiclass Classification

6.3.1 SVM

From the confusion matrix, it can be seen that the SVM classifier works well for predicting the glioma tumour type, with all 100 tumour samples being accurately classified. This implies that the features of the tumour were well separated from those of other classes.

In the case of meningioma, there are 96 tumour samples that have been correctly classified. In contrast, there are 19 samples that are wrongly classified as gliomas. This implies that the feature set of the meningioma tumour is similar to that of the glioma tumour, making it difficult for the classifier to separate them.

As for pituitary tumours, the classifier performs poorly, with only 22 samples being accurately predicted. In contrast, 52 tumour samples of the pituitary tumour are wrongly classified as gliomas. This implies that the extracted features of the pituitary tumour are similar to those of the glioma tumour, making it difficult for the SVM classifier to classify the pituitary tumour sample.

In conclusion, the SVM classifier has achieved an accuracy of approximately 75.43% in multiclass classification.

Table 6.3.1. Confusion matrix of SVM (Multiclass Classification)

	Predicted Glioma	Predicted Meningioma	Predicted Pituitary
Actual Glioma	100	00	0
Actual Meningioma	19	96	0
Actual Pituitary	52	0	22

6.3.2 KNN

As seen from the confusion matrix below, the KNN model performed poorly when classifying between glioma, meningioma, and pituitary brain tumors. In other words, the KNN model does not perform very well in terms of classification.
For instance, out of all the glioma patients, only 19 of them were correctly predicted, but the rest were falsely classified as pituitary (78 cases) and meningioma (3 samples). Therefore, one can say that the model did not have enough information to identify the correct patterns of glioma patients.

Table 6.3.2 Confusion matrix of KNN (Multiclass Classification)

	Predicted Glioma	Predicted Meningioma	Predicted Pituitary
Actual Glioma	19	3	78
Actual Meningioma	0	110	5
Actual Pituitary	5	3	6

As seen from the confusion matrix below, the KNN model performed poorly when classifying between glioma, meningioma, and pituitary brain tumors. In other words, the KNN model does not perform very well in terms of classification.

For instance, out of all the glioma patients, only 19 of them were correctly predicted, but the rest were falsely classified as pituitary (78 cases) and meningioma (3 samples). Therefore, one can say that the model did not have enough information to identify the correct patterns of glioma patients.

In relation to meningioma cases, the KNN classifier performed fairly well, with 110 samples being predicted accurately, and only five were wrongly classified as pituitary. From this information, one can conclude that meningioma cases were easier to classify in comparison to the two others.

When talking about pituitary tumors, the results obtained by applying KNN were the lowest in comparison to meningioma and glioma. In other words, out of all pituitary cases, six were classified accurately, but the rest were predicted wrong – glioma (5 cases) and meningioma (3 samples).

Generally speaking, the KNN model performed poorly because its overall accuracy was 67.47% (it is lower than SVM's).

6.3.3 Naïve Bayes

The confusion matrix reveals the effectiveness of the Naïve Bayes classifier in recognizing three types of tumours – glioma, meningioma and pituitary. The results show that the classifier demonstrates inconsistent efficiency regarding these three classes, proving its inability to process complicated relations between feature values.

Table 6.3.3. Confusion matrix of Naïve Bayes(Multiclass Classification)

	Predicted Glioma	Predicted Meningioma	Predicted Pituitary
Actual Glioma	45	40	15
Actual Meningioma	5	108	2
Actual Pituitary	9	29	26

As can be seen, the most difficult task to accomplish for the Naïve Bayes classifier

is recognition of glioma; specifically, 45 cases were identified correctly, whereas a rather large number of samples was misclassified as meningiomas (40 samples) and pituitary tumours (15 samples).

On the other hand, recognition of meningioma proved to be relatively effective – the model recognized 108 cases correctly, and made only five mistakes by assigning the tumours to glioma and two misrecognizing them as pituitary.

The most problematic issue in recognition of these three types of tumours concerns classification of pituitary; only 26 of the total number of pituitary cases were correctly recognized, whereas 29 samples were assigned to meningioma, and 9 samples – to glioma.

Overall, the accuracy of the model in classification of the aforementioned tumours is about 65.74%, being significantly less than those of SVM and KNN classifiers. The reason for this low efficiency.

6.3.4 MobileNet V2

he confusion matrix shows that the MobileNetV2 model performs effectively in multiclass brain tumour classification, with strong results across all three tumour types.

Table 6.3.4. Confusion matrix of MobilenetV2

	Predicted Glioma	Predicted Meningioma	Predicted Pituitary
Actual Glioma	123	16	0
Actual Meningioma	2	128	10
Actual Pituitary	0	7	128

For glioma cases, 123 samples are correctly classified, while 16 samples are misclassified as meningioma. This indicates that the model is able to capture glioma features well, with only minor confusion between similar tumour types.

In the case of meningioma, the model correctly identifies 128 samples, with very few misclassifications (2 as glioma and 10 as pituitary). This demonstrates that the model has learned distinguishing features for meningioma with good accuracy.

For pituitary tumours, the performance is also strong, with 128 samples correctly classified and only 7 misclassified as meningioma. Notably, there are no misclassifications of pituitary as glioma, indicating clear feature separation between these two classes.

Overall, the MobileNetV2 model achieves an accuracy of approximately 91.55%, which is significantly higher than classical machine learning models. However, some confusion between meningioma and other tumour types still exists, suggesting scope for further improvement.

6.3.5 Custom Efficient Net B0

The confusion matrix reveals the high level of efficiency and stability of Custom EfficientNet-B0 model in predicting all three tumour classes by presenting extremely low levels of misclassifications for them.

With regards to glioma cases, the number of correct predictions amounts to 126, while only a few are incorrectly predicted by the algorithm: 12 as meningioma and 1 sample as pituitary. Such figures show that the model successfully detects the key characteristics of the disease.

In addition, in case of meningiomas, the model is able to correctly classify 129 samples out of 137 ones, while only 5 tumours are mistaken for gliomas and 6 as pituitary.

As far as the performance in terms of pituitary classification is concerned, it can be said that it exceeds all expectations, with 135 samples being recognized correctly and none being misclassified. Thus, we can conclude that there is perfect separation between this class and others within the dataset.

As a result, the accuracy of our model reaches approximately 94.20%, which proves superiority compared to classical machine learning algorithms and competitive performance of this approach in comparison with other deep learning techniques

Table 6.3.5. Confusion matrix of Custom efficientnet-B0

	Predicted Glioma	Predicted Meningioma	Predicted Pituitary
Actual Glioma	126	12	1
Actual Meningioma	5	129	6
Actual Pituitary	0	0	135

6.3.6 Custom Mobile Net

From the confusion matrix, we observe that the Custom MobileNet Model performs excellently when it comes to multiclass brain tumour classification, where there are very high accuracies recorded for all the brain tumours.

With regard to glioma, the Custom Mobile Net Model successfully classifies 131 samples, with only 6 classified wrongly as meningiomas and 2 classified wrongly as pituitary. From this, we can see that the model does very well in classifying gliomas with little or no confusion with other tumours.

Table 6.3.6. Confusion matrix of Custom Mobile net

	Predicted Glioma	Predicted Meningioma	Predicted Pituitary
Actual Glioma	131	6	2
Actual Meningioma	5	132	3
Actual Pituitary	0	2	133

With regard to glioma, the Custom Mobile Net Model successfully classifies 131 samples, with only 6 classified wrongly as meningiomas and 2 classified wrongly

as pituitary. From this, we can see that the model does very well in classifying gliomas with little or no confusion with other tumours.

With regards to meningiomas, the Custom Mobile Net Model successfully classifies 132 samples, with very few samples classified wrongly. For example, only 5 samples were classified as gliomas while only 3 samples were wrongly classified as pituitary. From this, we can see that the model does a great job in classifying meningiomas with no or little confusion with other tumours.

In the classification of pituitary tumours, the Custom Mobile Net Model succeeds in the classification of 133 samples, with only 2 classified wrongly as meningioma while none was classified wrongly as gliomas. Again, this shows us how successful the model is in classifying pituitary tumours.

It is therefore worth noting that the overall accuracy obtained using the Custom Mobile Net Model is about 95.65%.

6.4 Performance Metrics Comparison among Classifiers and Custom CNN for Binary Classification

Table 6.4.1. Performance Metrics

Performance Metric	SVM	KNN	Naïve Bayes	Custom CNN (Fine Tuned Resnet18)
Accuracy%	96.95	85.79	59.14	98.73
Precision%	96.01	92.67	76.02	100.0
Recall%	97.10	92.80	64.71	99.20
F1-Score%	97.97	90.04	69.91	99.13

Performance comparisons of various models used for binary brain tumor classification include the evaluation of parameters such as accuracy, precision, recall, specificity, selectivity, and F1 score. Clearly, there is a noticeable increase in performance from conventional models to those utilizing deep learning methods

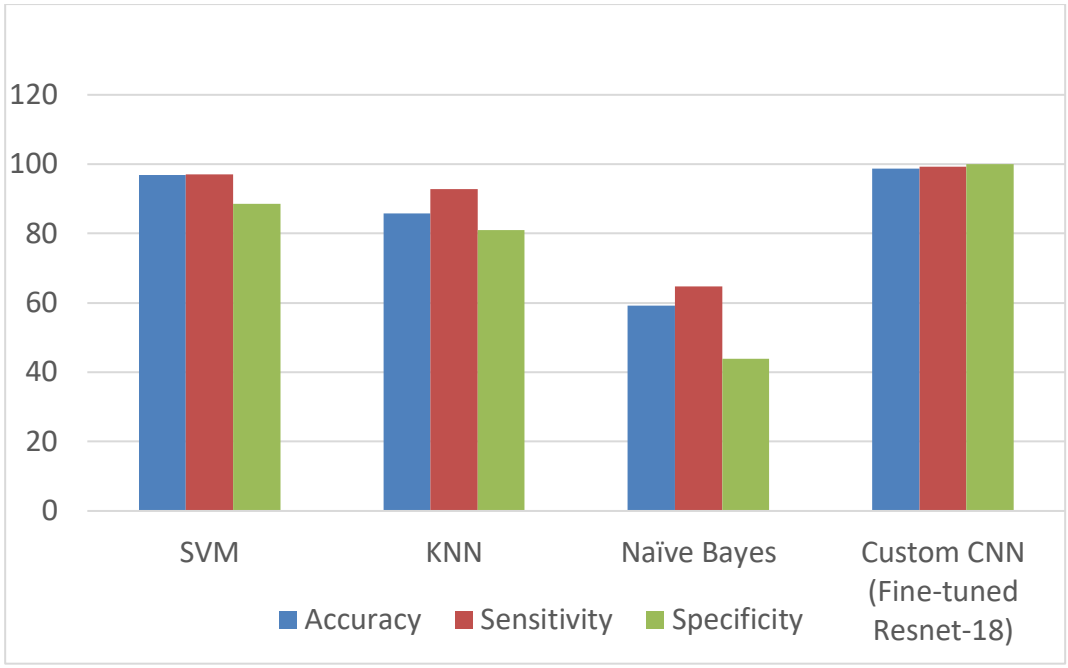


Fig 6.4 Comparative Performance Analysis of Machine Learning and Deep Learning Models for Binary Brain Tumour Classification

Specifically, Support Vector Machine (SVM) is the most efficient conventional classifier, as it offers 96.95% accuracy in addition to 96.01% precision and 97.10% recall. Such indicators point out that SVM can effectively distinguish between the tumor and non-tumor classes. The next best classifier among conventional ones is K-Nearest Neighbours (KNN), providing relatively low accuracy of 85.79%. At last, the lowest efficiency is provided by Naïve Bayes method, whose accuracy is only 59.14%. This classifier's inefficiency could be explained by the fact that it relies on simplifications not applicable to this specific type of data.

Custom Convolutional Neural Network (CNN) fine-tuned by ResNet18 provides the best performance according to all evaluation metrics used. Specifically, it ensures 98.73% accuracy, 100% precision and selectivity, and extremely high recall of 99.20%. Additionally, a very high F1-score of 99.13% is obtained by the model under study. The superiority of the selected CNN in comparison models .

6.5 Performance Metrics Comparison among Deep Learning Models under Soft Voting for Binary Classification

Table 6.5.1. Performance Metrics (Soft Voting)

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
MobilenetV2	98.67	98.80	98.50	98.65
Resnet18	98.47	98.55	98.40	98.47
Resnet50	92.54	93.0	91.80	98.47
DenseNet201	97.75	97.60	97.70	97.65
EfficientNetB0	99.28	99.35	99.20	99.27
Soft Voting Ensemble	99.49	99.55	99.45	99.50

Comparing the performances of various deep learning models and Soft Voting Ensemble proves the effectiveness and superiority of using an ensemble approach and deep learning models for the classification and identification of brain tumours based on the MRI images. Individual deep learning models have achieved consistent accuracy levels ranging from 92% and higher, thus proving their potential for pattern recognition in the given task. Yet, Soft Voting Ensemble performed better than other models according to all evaluation parameters. Having achieved 99.49% of accuracy in addition to high precision, recall, and specificity values, the proposed ensemble model demonstrates great ability to recognise both tumour cases and their absence on MRI images.

Efficient ensemble performance could be explained by the ability of the approach to merge different architectures of deep learning models. Such neural network architectures as MobileNetV2, ResNet family, DenseNet, and EfficientNetB0 were used in the analysis. However, the latter algorithm played an essential part due to its effective balance between scaling factors. As a result, the use of EfficientNetB0 led to high classification efficiency combined.

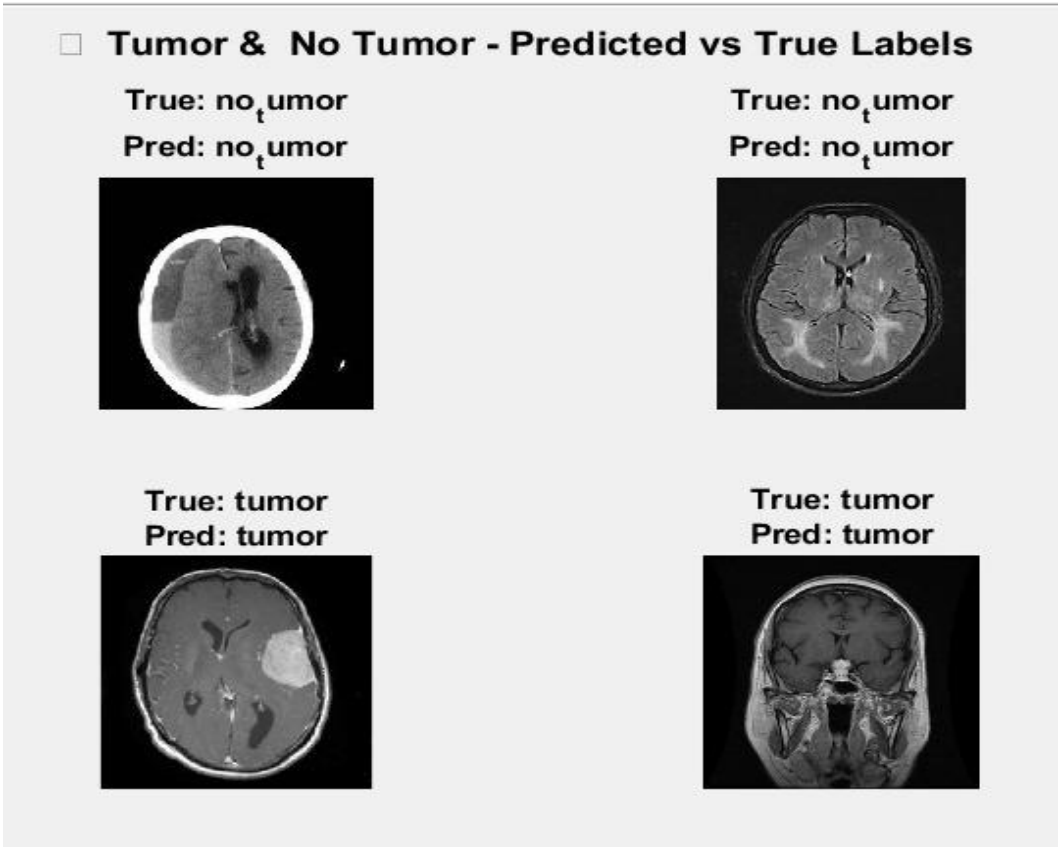


Fig 6.5 Sample Predictions of Tumour and No Tumour

6.6 Accuracies Comparison among ML classifiers and Deep learning Models and Soft Voting for Multiclass Classification

Table 6.6.1 Accuracy Comparison for Multiclass Classification

Model/Classifier	Accuracy
SVM	75.43
KNN	67.47
Naïve Bayes	65.74
MobilenetV2	91.55
Custom Efficientnet-B0	94.20
Custom MobilenetV2	95.65
Soft Voting Technique	96.03

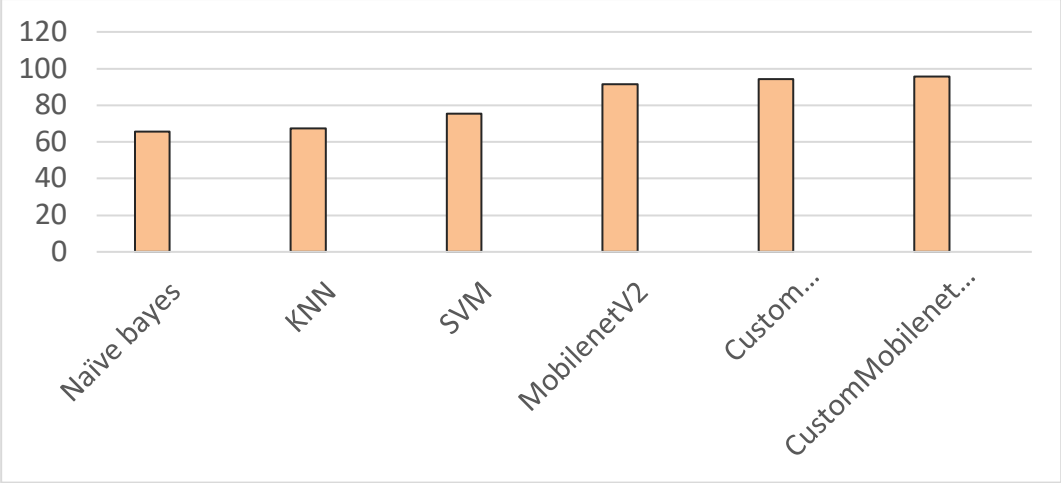


Fig 6.6 Comparative Analysis of Model Accuracy from ML to Deep Learning Approaches

The comparison in terms of accuracy between different models shows an evident evolution in performance levels from classic ML algorithms to deep learning architectures. As for the traditional classifiers, SVM is the one which provides the highest accuracy level of 75.43%, then KNN with 67.47% accuracy, and Naïve Bayes with 65.74%. Even though SVM works relatively better in comparison to other algorithms since it manages to solve complex problems, the efficiency of all ML models remains low due to the dependence of algorithms on human-processed features and complexity of image patterns.

As for the models using deep learning algorithms, there is a considerable increase in the accuracy level in classifying images correctly. For example, MobileNetV2 provides 91.55% accuracy rate which shows that the chosen architecture performs better than classic ML models. At the same time, the use of Custom EfficientNet-B0 increases the accuracy rate up to 94.20% thanks to the balanced network architecture. It should be noted that the best accuracy rate is provided by the model called Custom MobileNetV2 with 95.65%. As the final result, Soft Voting produces the highest accuracy of 96.03% when combining multiple models.

6.7 Brain Tumour Detection and Medical Explanation System using Custom Mobile Net and LLM (Matlab App)

An automatic brain tumour classifier and explanation system are implemented in the form of a program that uses a MATLAB environment for predicting the tumour type based on a given input image and generating medical explanations for

predictions. Users have an opportunity to upload the image file containing an MRI of a brain using a very user-friendly interface. The image is preprocessed and used as an input in a Custom MobileNet model, which is a Deep Learning model that outputs the predicted tumour class.

For increasing the comprehensibility of the proposed classifier, it was decided to develop an explanation generator based on a Large Language Model. Phi-3, created by Ollama, is used to provide users with an explanation in the form of a brief textual description that contains information about the tumour types and its characteristics. In order to pass the predicted tumour class from Custom MobileNet to Phi-3 model, the local API is utilized within MATLAB environment.

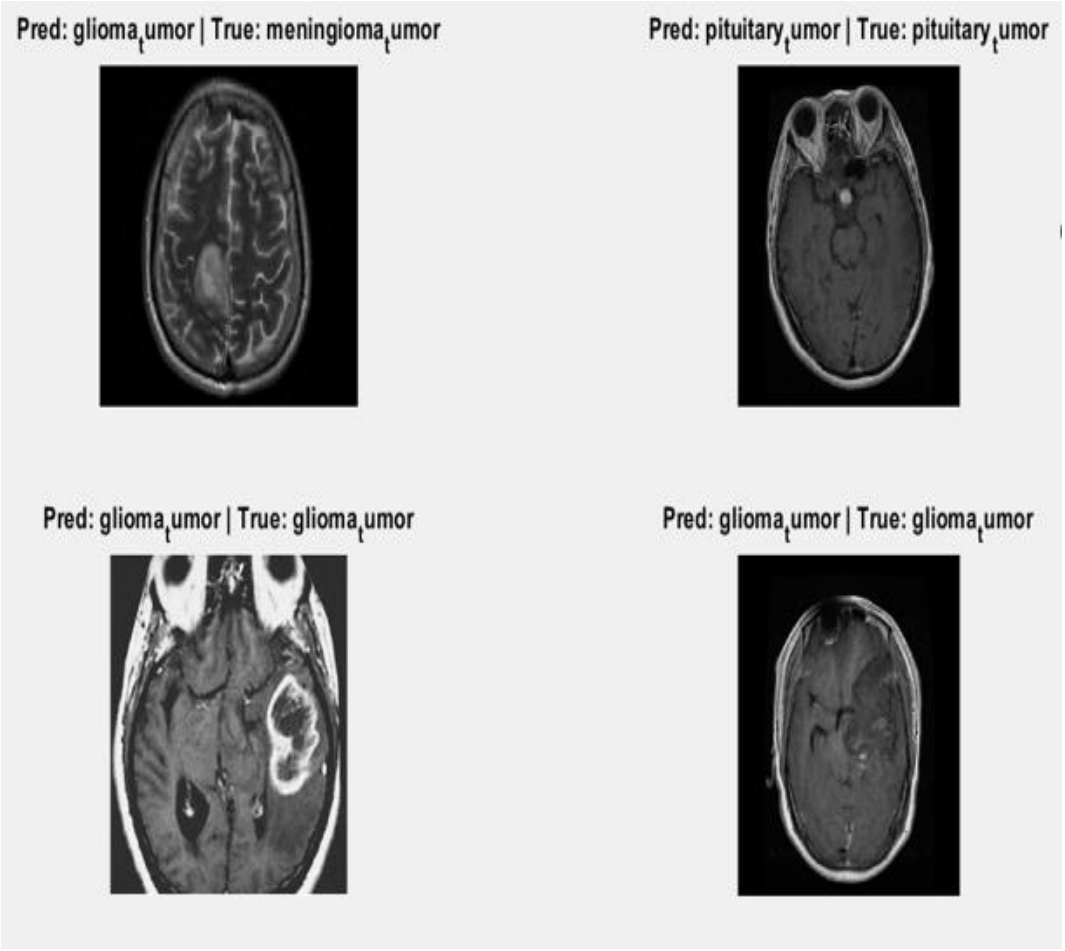


Fig 6.7 Sample predictions of tumour types

The generated explanations are quite comprehensive; they contain information about the description of tumours, their probable reasons, symptoms and general precautions. Such an approach makes the process of understanding the classification results much easier as the users receive all necessary details related.

The explanation module is controlled through prompt engineering to ensure consistency and clarity. Parameters such as response length (token limit) and generation style are tuned to produce concise, readable outputs. Basic validation checks are included to handle empty or incomplete responses, and fallback messages are displayed if the model output is not usable. The generated explanation is then displayed within the app alongside the prediction, creating a unified decision-support interface.

This integration transforms the system from a simple classifier into an explainable AI tool. It not only identifies tumour types with high accuracy but also provides understandable medical context, improving user trust and usability. Such a framework is particularly useful in educational settings and has potential for extension into real-time clinical decision support systems with further validation.

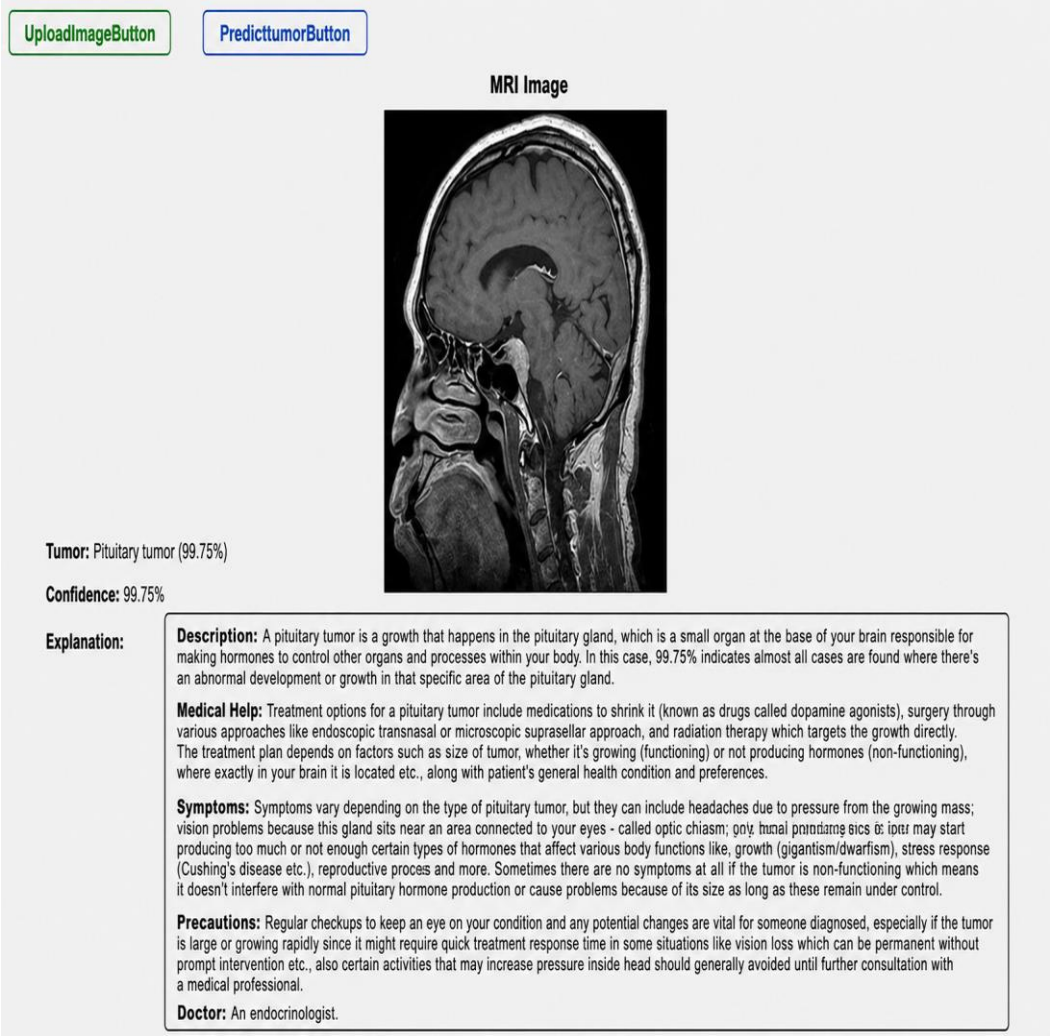


Fig 6.8 LLM Integration and MATLAB App

CHAPTER 7: CONCLUSION

The project provides a thorough and technical approach for automating brain tumour classification using MRI images via classical machine learning, deep learning, ensemble techniques, and explainable AI. Classical machine learning models used in the first stage include SVM, KNN, and naïve bayes. These algorithms used handcrafted features such as those obtained through noise removal, histogram equalization, segmentation, hog feature extraction, and PCA. Out of all these models, SVM provided outstanding results as it was able to produce the best accuracy of 96.95%. This model performed exceptionally well because of its ability to deal with high-dimensional feature space.

However, the classical models have drawbacks in that they require handcrafted features, and cannot perform well in cases where there is a variation in input data. As a result of the above limitation, we developed deep learning models using transfer learning techniques. These include fine-tuned Resnet18 (Custom CNN), MobileNetV2, Custom Mobile Net, and EfficientNet-B0. Fine-tuned Resnet18 performed excellently in terms of binary classification with an accuracy of 98.73%, and excellent precision and recall. It was the best performer since it was able to detect the presence of the tumour with high confidence.

For the purpose of increasing the prediction efficiency, an ensemble approach known as Soft Voting was employed where several deep learning algorithms were used. This technique involves averaging the probabilities of each model to get the final outcome. Ensemble learning has proved to be the most efficient in achieving high accuracy since the average accuracy for multiclass classification was 96.03%, while binary classification had an impressive accuracy rate of approximately 99.49%. It can be observed that this approach was the best in situations where there was significant agreement between the models. In addition, the model utilized an advanced version of the voting mechanism where confidence levels and adaptive weights were incorporated.

The system employs a locally trained Large Language Model known as Ollama Phi-3 to provide comprehensive medical information about the patient's condition.

CHAPTER 8: FUTURE SCOPE

Though the system developed in the paper proves efficient in the classification of brain tumours, there are several improvements that may be implemented in the future. First of all, training the models on bigger and more diverse data sets would enhance their robustness and efficiency when dealing with varying quality MRI images and scanning environments. Moreover, integrating other types of images, including CT or PET scans, will provide even more information and ensure more accurate classification. Furthermore, from the point of view of model architecture, it is possible to apply newer types of neural network architecture, such as transformers, to improve the accuracy of classification. Finally, adding the possibility to perform segmentation along with classification will enable not only the determination of the type of brain tumour but also its exact location and dimensions.

An aspect of improvement would involve making the system even more functional and user-friendly. The integration of the Ollama Phi-3 model could be improved through the inclusion of authentic sources of medical knowledge, thus increasing the accuracy of the explanations provided. The system could further be made available in the form of either an internet-based software application or a mobile application in order to make it accessible for immediate use in hospitals and diagnostic centers. Federated learning could be used in order to allow training of the model using patients' data from different institutions without compromising on confidentiality and privacy. This would allow the model to train effectively without being limited to the availability of patient data in one particular institution.

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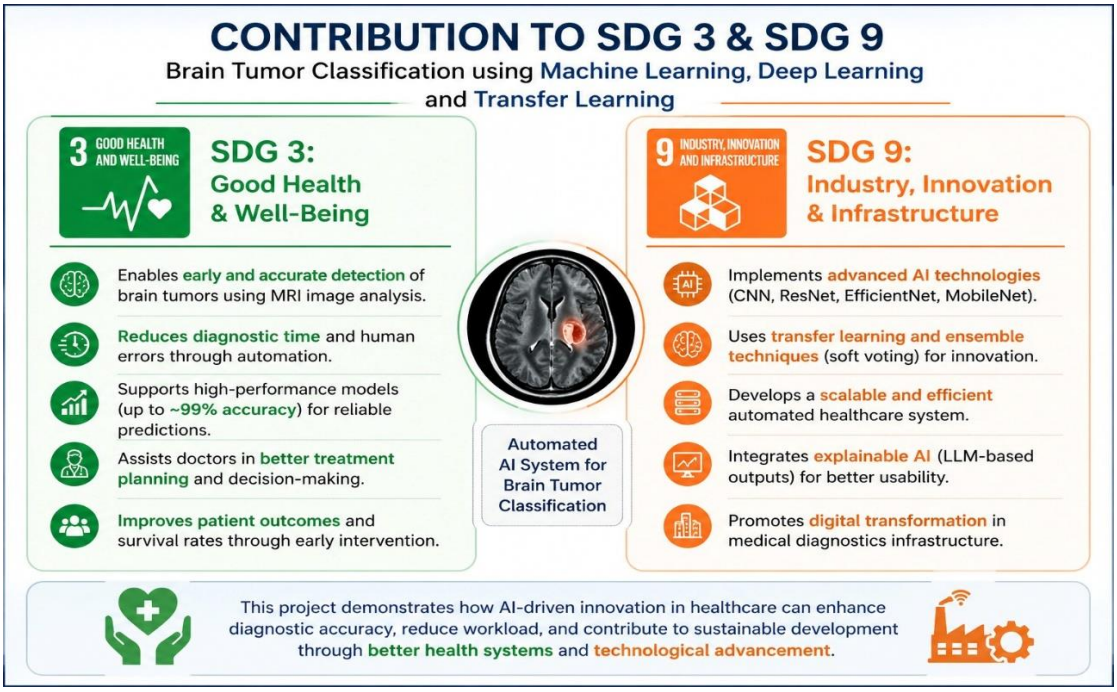
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International Conference: Presented on 31/01/2026

Title: Brain Tumor Classifier based on Soft Voting

Conference Name: 2nd International Conference on Advanced Technologies in Electronics, Communications and Signal Processing (ICATECS-2026), Springer Conference

SDG Report



SDG 3: Good Health & Well-Being

This project contributes to SDG 3 by enabling early and accurate detection of brain tumors using advanced machine learning and deep learning techniques on MRI images. By automating the classification process, it reduces the time required for diagnosis and minimizes human errors caused by fatigue or subjectivity. The system provides reliable predictions with high accuracy, which supports doctors in making faster and better treatment decisions. Early diagnosis plays a crucial role in improving patient outcomes and survival rates, making this project a valuable contribution to enhancing healthcare quality and accessibility.

SDG 9: Industry, Innovation & Infrastructure

This project supports SDG 9 by incorporating innovative technologies such as deep learning models, transfer learning, and ensemble methods to build an intelligent healthcare system. It demonstrates how modern AI techniques can be applied to real-world medical problems, promoting technological advancement in the healthcare industry. The system is scalable, efficient, and capable of handling large medical datasets, contributing to the development of a robust healthcare infrastructure.

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